

AI Powered Workplace Monitoring System

A PROJECT REPORT

Submitted by

B V TARUN DEEP REDDY [RA2211003011065]

K BHARATH ROYAL [RA2211003011082]

Under the Guidance of

(Dr.G.PADMAPRIYA)

(Associate Professor, Department of Computing Technologies)

in partial fulfillment of the requirements for the degree of

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE AND ENGINEERING



DEPARTMENT OF COMPUTING TECHNOLOGIES

COLLEGE OF ENGINEERING AND TECHNOLOGY

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

KATTANKULATHUR- 603 203

MAY 2026

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY KATTANKULATHUR – 603 203

BONAFIDE CERTIFICATE

Certified that 21CSP401L/21CSP402L – Major Project [21CSP403L-Internship] report titled “**AI POWERED WORKPLACE MONITORING SYSTEM**” is the bonafide work of “**B.V TARUN DEEP REDDY [RA2211003011065], K BHARATH ROYAL [RA2211003011082]**” who carried out the project work under my supervision. Certified further, that to the best of my knowledge the work reported herein does not form any other project report or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

SIGNATURE

Dr.G. Padmapriya

SUPERVISOR

Associate Professor
Department of
Computing Technologies

SIGNATURE

Dr.G. Niranjana

PROFESSOR & HEAD

Department of
Computing Technologies

EXAMINER 1

EXAMINER 2

To be completed by the student for all assessments

Degree/ Course : **B.TECH/ Computer Science and Engineering**
Student Name : **B.V TARUN DEEP REDDY, K BHARATH ROYAL**
Registration Number : **RA2211003011065, RA2211003011082**
Title of Work : **AI POWERED WORKPLACE MONITORING SYSTEM**

I / We hereby certify that this assessment compiles with the University's Rules and Regulations relating to academic misconduct and plagiarism, as listed in the University Website, Regulations, and the Education Committee guidelines.

I / We confirm that all the work contained in this assessment is my / our own except where indicated, and that I / We have met the following conditions:

- Clearly referenced / listed all sources as appropriate
- Referenced and put in inverted commas all quoted text (from books, web, etc)
- Given the sources of all pictures, data etc. that are not my own
- Not made any use of the report(s) or essay(s) of any other student(s) either past or present
- Acknowledged in appropriate places any help that I have received from others (e.g. fellow students, technicians, statisticians, external sources)
- Compiled with any other plagiarism criteria specified in the Course handbook / University website

I / We understand that any false claim for this work will be penalized in accordance with the University policies and regulations.

DECLARATION:

I am aware of and understand the University's policy on Academic misconduct and plagiarism and I certify that this assessment is my / our own work, except where indicated by referring, and that I have followed the good academic practices noted above.

B.V TARUN DEEP REDDY

K BHARATH ROYAL

ACKNOWLEDGEMENTS

We express our humble gratitude to **Dr. C. Muthamizhchelvan**, Vice-Chancellor, SRM Institute of Science and Technology, for the facilities extended for the project work and his continued support.

We extend our sincere thanks to **Dr. Leenus Jesu Martin M**, Dean-CET, SRM Institute of Science and Technology, for his invaluable support.

We wish to thank **Dr. Revathi Venkataraman**, Professor and Chairperson, School of Computing, SRM Institute of Science and Technology, for her support throughout the project work.

We encompass our sincere thanks to, **Dr. M. Pushpalatha**, Professor and Associate Chairperson - CS, School of Computing and **Dr. C Lakshmi**, Professor and Associate Chairperson -AI, School of Computing, SRM Institute of Science and Technology, for their invaluable support.

We are incredibly grateful to our Head of the Department, **Dr. Niranjana G**, Designation of Computing Technologies, SRM Institute of Science and Technology, for her suggestions and encouragement at all the stages of the project work.

We want to convey our thanks to our Project Coordinators, Panel Head, and Panel Members Department of Computational Technologies, SRM Institute of Science and Technology, for their inputs during the project reviews and support.

We register our immeasurable thanks to our Faculty Advisor, **Dr. Kandan M**, Department of Computing Technologies, SRM Institute of Science and Technology, for leading and helping us to complete our course.

Our inexpressible respect and thanks to our guide, **Dr.G.Padmapriya**, Department of Computing Technologies, SRM Institute of Science and Technology, for providing us with an opportunity to pursue our project under his / her mentorship. He / She provided us with the freedom and support to explore the research topics of our interest. His / Her passion for solving problems and making a difference in the world has always been inspiring.

We sincerely thank all the staff members of Department of Computing Technologies, School of Computing, S.R.M Institute of Science and Technology, for their help during our project. Finally, we would like to thank our parents, family members, and friends for their unconditional love, constant support and encouragement.

B V TARUN DEEP REDDY [RA2211003011065]

K BHARATH ROYAL [RA2211003011082]

ABSTRACT

The rapid advancement of Artificial Intelligence (AI) and computer vision technologies has facilitated the development of intelligent systems for automating workplace monitoring and attendance management. Traditional workplace monitoring and attendance systems, such as manual registers, RFID cards, and biometric systems, suffer from various inefficiencies, limitations, and susceptibility to manipulation. In this research paper, the proposed solution consists of an AI-based employee monitoring system that uses artificial intelligence and computer vision for automatic detection and monitoring of employees in a workplace setting. For improved accuracy in facial recognition, the images are normalized using grayscale normalization and histogram equalization methods. Moreover, the solution also involves the development of an administrator authentication module that offers secure access to monitoring sessions. The monitoring module monitors and detects employees from live camera feed and records their presence based on the time they spend in the camera's field of view. A time-based system is also used to verify the presence of employees, which requires them to remain visible for 80 percent of the session. Based on experimental results, the proposed solution successfully delivers accurate real-time monitoring of employees. Furthermore, the framework reduces the need for manual supervision and improves the reliability of attendance tracking under diverse working conditions.

TABLE OF CONTENTS

ABSTRACT		v
TABLE OF CONTENTS		vi
LIST OF FIGURES		vii
LIST OF TABLES		viii
ABBREVIATIONS		ix
CHAPTER NO.	TITLE	PAGE NO.
1	INTRODUCTION	1
1.1	Introduction to Project	1
1.2	Problem Statement	1
1.3	Motivation	2
1.4	Sustainable Development Goal of the Project	3
2	LITERATURE SURVEY	4
2.1	Overview of the Research Area	4
2.2	Existing Models and Frameworks	4
2.3	Limitations Identified from Literature Survey (Research Gaps)	5
2.4	Research Objectives	6
2.5	Product Backlog (Key user stories with Desired outcomes)	6
2.6	Plan of Action (Project Road Map)	7
3	SPRINT PLANNING AND EXECUTION METHODOLOGY	9
3.1	SPRINT I - Dataset Preparation & Baseline Model Development	9
3.1.1	Objectives with user stories of Sprint I	9
3.1.2	Functional Document	11
3.1.3	Architecture Document	14
3.1.4	Outcome of Objectives / Result Analysis	18
3.1.5	Sprint Retrospective	20
3.2	SPRINT II - Fairness Enhancement by CSDG & Selective Fine-Tuning	21
3.2.1	Objectives with user stories of Sprint II	21
3.2.2	Functional Document	23
3.2.3	Architecture Document	25
3.2.4	Outcome of objectives/ Result Analysis	26
3.2.5	Sprint Retrospective	28

3.3	SPRINT III - Robustness Enhancement & Deployment Readiness	29
3.3.1	Objectives with user stories of Sprint III	29
3.3.2	Functional Document	30
3.3.3	Architecture Document	32
3.3.4	Outcome of objectives/ Result Analysis	35
3.3.5	Sprint Retrospective	36
4	RESULTS AND DISCUSSIONS	37
4.1	Project Outcomes (Performance Evaluation, Comparisons, Testing Results)	37
5	CONCLUSION AND FUTURE ENHANCEMENT	42
	REFERENCES	44
	APPENDIX	
A	CODING	46
B	CONFERENCE PUBLICATION	82
C	JOURNAL PUBLICATION	84
D	PLAGIARISM REPORT	85

LIST OF FIGURES

CHAPTER NO.	TITLE	PAGE NO.
3.1	MS Planner activities for Sprint 1 along with the User Story	10
3.2	MS Planner activities for Sprint 1 along with the User Story	11
3.3	Architecture Diagram for the Proposed Research Framework	14
3.4	Use case diagram for the Proposed Research Framework	15
3.5	Dataset Audit and Analysis Results during Sprint I	19
3.6	Underrepresented Classes & the Imbalance Ratio in the dataset	19
3.7	Ordered Splitting of the Dataset for Model Development	19
3.8	Sprint Retrospective for Sprint I	20
3.9	MS Planner Activities for Sprint II	22
3.10	MS Planner Activities for Sprint II	22
3.11	MS Planner Activities for Sprint II	23
3.12	Class Diagram for the Proposed Research Framework	26
3.13	Successful Completion of the CSDG Phase	27
3.14	Validation of CSDG Phase	27
3.15	Initial output of the SFT Classifier	28
3.16	Sprint Retrospective for Sprint II	28
3.17	MS Planner Activities for Sprint III	30
3.18	Component Diagram of the Proposed Research Framework	33
3.19	Sequence Diagram of the Proposed Research Framework	34
3.20	Evaluation During the Robustness Tests	35
3.21	List of Robustness Tests for Real – World Suitability	36
3.22	Sprint Retrospective for Sprint III	36
4.1	Crop Parity Gap Comparison Across Models	39
4.2	Ablation Study Across Different Models	39
4.3	Robustness Curves Across Multiple Corruptions	41
B.1	Acceptance Notification from ICCIET 2026	82
B.2	Certificate of Participation for Sai Teja Mandava	83
B.3	Certificate of Participation for Sree Charanya Dantuluri	83
C.1	Copyright form for the Research Paper	84
D.1	Similarity Index for the Project Report	85
D.2	Similarity Index for the Project Report	85

ABBREVIATIONS

AI	Artificial Intelligence
CV	Computer Vision
FRS	Face Recognition System
FD	Face Detection
LBPH	Local Binary Pattern Histogram
HC	Haar Cascade
RTM	Real-Time Monitoring
WMS	Workplace Monitoring System
AMS	Attendance Management System
ADM	Administrator Module
DB	Database
GUI	Graphical User Interface
API	Application Programming Interface
FPS	Frames Per Second
ROI	Region of Interest
RGB	Red Green Blue
GS	Grayscale
TP	True Positive
TN	True Negative
FP	False Positive
FN	False Negative
ACC	Accuracy
LAT	Latency
PR	Presence Ratio
ST	Session Time
PT	Presence Time

CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION TO PROJECT

With the development of Artificial Intelligence (AI) technology, conventional employee tracking and attendance solutions have been developed into advanced systems. The most commonly used traditional technologies include manual logs, RFID chips, and biometric identification techniques. All of these traditional tracking techniques suffer from problems including inefficiency, inaccuracy, and manipulability. As organizations expand and become more complex, these traditional techniques are increasingly becoming impractical and cumbersome to use. There is an urgent need for a smart solution that is capable of continuously monitoring and recording the attendance of employees in the workplace.

In particular, this system employs a camera-based solution, wherein the face detection process is accomplished using the Haar Cascade algorithm, while the face recognition step involves the use of the LBPH model. Preprocessing steps, including grayscale processing and histogram equalization, will be conducted to enhance accuracy during both face detection and recognition procedures. This system automatically monitors live video feed to detect employees and logs their attendance based on how long they stay within the monitored environment. It also includes an authentication process for administrators and a time-based attendance assessment procedure, wherein an individual must be detected in 80 percent of the session to be counted as present.

1.2 PROBLEM STATEMENT

There are some problems associated with using attendance and monitoring solutions in practical settings, even when there are several different kinds of systems available today. For example, using traditional approaches such as manual registration, RFID cards, and biometric systems is not very efficient, accurate, and cannot prevent manipulation by workers. Moreover, there may be such problems as variations in illumination, orientation, occlusion (for instance, wearing masks and hats), and lack of camera coverage when it comes to current AI systems, and all of these factors may result in a lower efficiency of facial recognition. In addition, many modern AI solutions do not constantly check if employees are present on their workplaces.

In addition, most of the current techniques have mainly concentrated on the precision of detection and recognition without taking into consideration continuous real-time monitoring and practical usability. These techniques do not provide security for access control, efficient processing, and proper methods for evaluating attendance. Attendance will be marked correctly by some of these techniques regardless of whether the employee was actually in attendance for a long enough period or not. Scalability and flexibility are also problems because most systems cannot efficiently handle different work settings. This necessitates the design of a robust and smart system capable of detecting and recognizing employees, monitoring their attendance in real time, securing their access, and fairly evaluating their attendance based on time.

1.3 MOTIVATION

The underlying reason for creating an artificial intelligence-based system for monitoring and managing the attendance of employees is based on the need for enhancing the effectiveness of employee monitoring systems. While conventional means of employee tracking and monitoring are not only cumbersome but also prone to errors, these tools cannot be applied for the contemporary world because of their ineffectiveness. Although there are a number of AI-based systems already being used, a large number of them face issues when performing in the real world due to factors like changing light conditions, varying face positions, and dynamic workplaces.

The other main factor is to make sure that there is fairness and integrity in terms of attendance recording. Many of the currently existing methods record the attendance without taking into account how long someone was present, making the data unreliable. In this case, the project intends to solve this problem by implementing a time-based method of assessing attendance. Moreover, the provision of authentication in terms of administrators' security ensures proper control and minimizes any form of intrusion. All these factors play a significant role in motivating this project towards developing an effective, intelligent monitoring system.

1.4 SUSTAINABLE DEVELOPMENT GOAL OF THE PROJECT

This particular project contributes to several Sustainable Development Goals (SDGs) suggested by the UN. Specifically, it is aligned with SDG 9: Industry, Innovation, and

Infrastructure and with SDG 8: Decent Work and Economic Growth. With the introduction of a new monitoring tool powered by AI, a company is capable of applying innovative technologies such as computer vision and machine learning into practice to enhance their manufacturing process and labor management.

As a result, this technology would allow firms to operate more efficiently without spending additional human effort and resources. Furthermore, the project helps to achieve Goal 12: Responsible Consumption and Production because it ensures minimal wastage of resources in terms of the current conventional attendance methods. For instance, the adoption of the time-based attendance assessment mechanism ensures that there is fairness and accuracy during the process, which motivates the workers to be accountable and responsible at their jobs. Ultimately, the system ensures minimal wastage of resources and ensures a sustainable and ethical business environment.

CHAPTER 2

LITERATURE SURVEY

2.1 OVERVIEW OF THE RESEARCH AREA

Inclusion of the technology of artificial intelligence and computer vision has brought a huge change into the field of monitoring and tracking the attendance of employees in the workplace. It is one of the critical concerns for all organizations nowadays since efficiency can be achieved only if employees' attendance is monitored properly. Manually tracking the attendance of employees along with RFID cards and biometrics has proved to be a failure because of the probability of human error and manipulation. In recent years, as a result of rapid advancements in AI, scientists and researchers have focused on developing a smarter system with the help of cameras.

The detection and recognition techniques play an important role in the construction of the abovementioned surveillance systems. The Haar Cascade technique is normally used in detecting the face since it gives fast output, while LBPH technique is normally used for recognizing the face through the consideration of the facial features. If the image processing techniques, including the transformation of the images into grayscale images and the histogram equalization, are integrated into intelligent surveillance systems, then the performance of the system would be optimal despite the prevailing environment. Even though the present systems have demonstrated excellent performance in laboratory conditions, there are some challenges such as varying light and obstacles that make them inaccurate in practice.

2.2 EXISTING MODELS AND FRAMEWORKS

Some of the models and theories formulated for workplace monitoring and attendance management through Artificial Intelligence and computer vision technology are as follows. Previously, most of the workplace surveillance systems had a tendency of depending on manual processes, RFID cards, and biometric technologies that were not only error-prone but also manipulative. With the advent of computer vision technology, researchers started to look into other options where surveillance could be achieved automatically using cameras and face detection technology. Some of the commonly used algorithms in this category include the

Eigenface algorithm, the Fisherface algorithm, and the Local Binary Pattern Histogram (LBPH).

There are other technologies that have been used; some include the application of pre-processing and object detection. One of the popular methods that have been used for face detection in videos is the Haar Cascade algorithm. Pre-processing involves converting an image into greyscale, equalizing histograms, and removing any form of noise from the image to increase accuracy and efficiency. Modern technology is equipped with features such as video surveillance, attendance recording, and reporting. Some challenges associated with modern technology include detecting multiple faces at once and handling occlusions and lighting variations.

2.3 LIMITATIONS IDENTIFIED FROM LITERATURE SURVEY (RESEARCH GAPS)

Despite the numerous advancements made in the development of face recognition technologies that can be applied in the workplace setting, there are still some limitations that should be addressed before implementing such facial recognition technology in the workplace. From the literature review conducted, it has been noted that among the major limitations associated with the face recognition system is that these facial recognition technologies rely on controlled settings. There are facial recognition technologies that perform well in a controlled environment but do not perform efficiently when faced with varying lighting conditions and camera angle, as well as other factors like partial mask occlusions.

Another important drawback is the absence of continual and precise measurement of attendance of an employee. The majority of currently developed software solutions pay attention only to identification and detection of particular individuals but ignore tracking their attendance during the whole period of monitoring. It often results in incorrect logging of employees' attendance since they might appear in the workplace for a very brief period. Also, most of the systems fail to provide sufficient algorithms and tools for supporting several users at once and maintaining consistent user recognition.

Moreover, the existing technologies fail to tackle critical issues like those of fairness, transparency, and security, concentrating more on the accuracy of recognition. It is true that a vast majority of these systems lack appropriate means of authentication, thus being prone to

any sort of attacks. Moreover, there exists little concentration on scalability issues and effective integration of these systems into existing office practices. Clearly, there is a demand for more robust solutions which will guarantee continuous surveillance, fair assessment of attendance and reliable access to resources.

2.4 RESEARCH OBJECTIVES

The primary goal of the project is to design and develop an artificial intelligence-driven system for monitoring the workplace environment to detect and identify employees through computer vision technology in an accurate, effective, and timely manner. Whereas existing solutions concentrate on mere employee attendance recording, this project will help address issues related to inefficiency, opacity, and vulnerability to manipulation. The proposed solution will employ camera technology and will leverage the Haar Cascade algorithm for detecting faces and the Local Binary Pattern Histogram technique for identifying the same. Furthermore, efforts will be made to improve the performance of the system by employing image processing methods such as grayscaling and histogram equalization.

In addition to the above, another important goal of the proposed study is to provide a continuous monitoring system capable of measuring the period of the employee's presence and not just one moment when they are detected. An attendance assessment procedure based on the time period is used. To be considered present, an employee should be visible during a certain percent (80%) of the monitoring process. It ensures the correctness and authenticity of attendance monitoring. Moreover, the proposed system will include a reliable authentication module, which will require entering administrative credentials before proceeding with any monitoring operations.

2.5 PRODUCT BACKLOG (KEY USER STORIES WITH DESIRED OUTCOMES)

A Product Backlog is the most crucial aspects of the proposed system based on the user-centered view. For instance, for the proposed AI-driven workplace monitoring system, some of the main user stories include the ability of an admin to log into the system and easily start a monitoring session. The system should have a straightforward and easy-to-use interface where one can determine the time for monitoring and see live results. Additionally, another key aspect

is the automatic detection and identification of employees in the workspace through the use of the camera and tracking them without any manual efforts.

Moreover, the system must have the ability to work efficiently in practical scenarios, which include changes in lighting, presence of several people within the same camera view, and diverse workplace environments. An important user story includes the requirement for the system to constantly track the presence of employees and measure their attendance according to the set time threshold (80% of session time, for example). Also, the system must protect user data privacy, automatically produce attendance records, and deliver reliable results despite the adverse conditions. The system must also be lightweight and scalable to make it applicable to varied organizational settings.

2.6 PLAN OF ACTION (PROJECT ROAD MAP)

Methodology that would be employed in executing this research will be carried out in a step-by-step manner to ensure effective development and testing of the artificial intelligence technology-based workplace surveillance system. First of all, there would be the process of data acquisition and preprocessing, in which the images of faces of workers would be captured using a camera system. Data would be processed through methods such as converting them into grayscale, normalization, and histogram equalization. This would enable effective processing of the dataset. Quality of the input data plays an important role in the functioning of the face recognition system. The third step would involve training of the algorithm through the use of Local Binary Pattern Histogram (LBPH).

The second step in our design involves transitioning the system to real-time monitoring mode. In this step, video frames obtained in real-time from the camera are analyzed. The faces are detected using the Haar Cascade algorithm, whereas the LBPH model is employed for recognition. It is worth noting that the system will track the presence of each person after successful recognition to ensure uninterrupted monitoring. To establish the presence of each individual, a time-based attendance calculation method is utilized, requiring that each person should be captured for 80 percent of the total session time. Lastly, attendance reports are generated, and performance evaluation takes place through an assessment of recognition accuracy, real-time performance, and reliability in different conditions, including lighting variation, among others.

Moreover, apart from the functionalities discussed above, other sophisticated features could be incorporated into the system that include multiple cameras for better coverage, which would increase the precision of attendance tracking in the workplace. In order to improve scalability, the cloud-based storage and real-time analytics can be considered. This way, the collected attendance data can be accessed remotely and even analyzed in the cloud. Additionally, the system can be supplemented with behavioral tracking capabilities for identifying any abnormal activity in the organization. Regular updating of the system models with new data would improve adaptability.

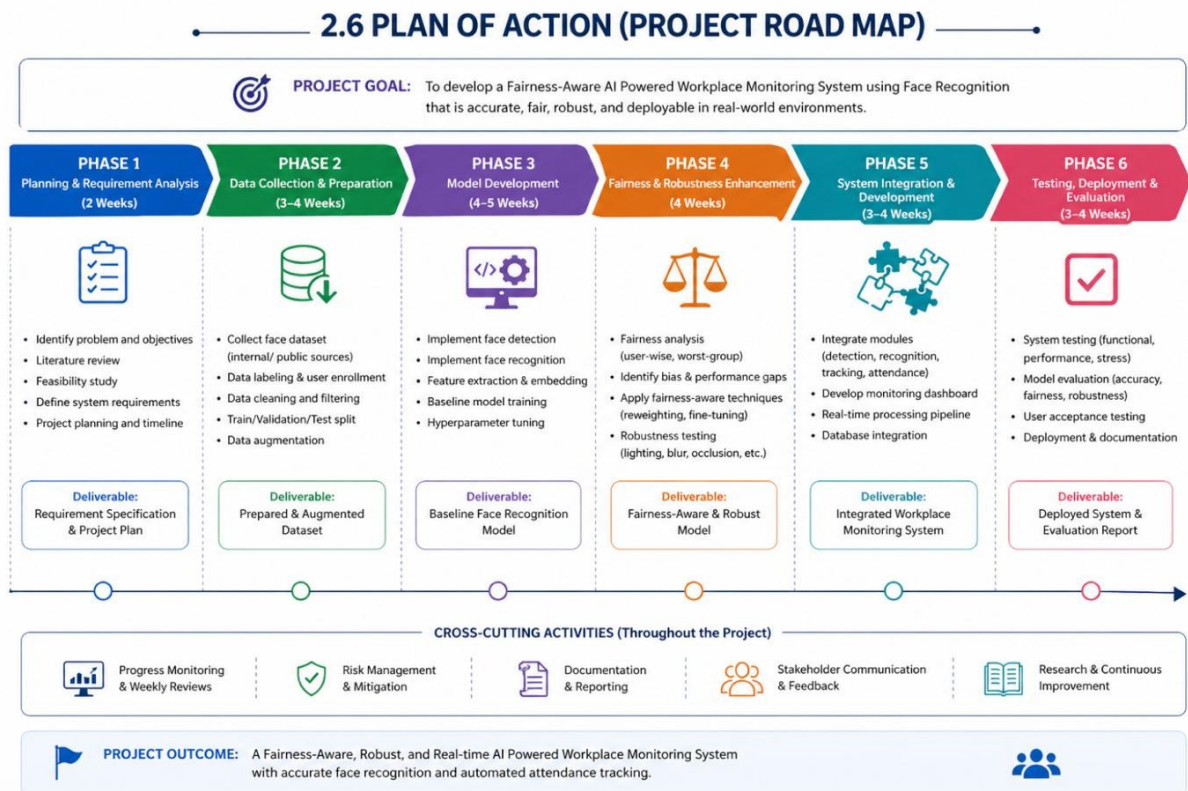


Fig 1. PLAN OF ACTION (PROJECT ROAD MAP)

CHAPTER 3

SPRINT PLANNING AND EXECUTION METHODOLOGY

3.1 SPRINT I – Dataset Preparation & Baseline Model Development

3.1.1 Objectives With User Stories Of Sprint I

During Sprint I, the main emphasis is on developing the fundamental components of the AI-driven workplace monitoring system. The first step is to develop the pipeline necessary for face detection and recognition. In this process, the system captures images of users' faces through a webcam. These images contain faces in different orientations, including faces that are straight, turned left, and turned right. Preprocessing is done next, where the images are converted to grayscale format and resized for uniformity. Further, they undergo normalization and histogram equalization to improve their quality.

S.NO	Detailed User Stories
User Story #1	The user story relates to an administrator or a system operator in charge of organizing user data. The administrator needs to be capable of capturing, storing and organizing user facial images in a structured dataset. The system will make sure that proper management of the dataset is achieved. This ensures that the quality of the training dataset is maintained hence increased recognition accuracy.
User Story #2	This user story relates to the developer or system engineer in charge of pre-processing of the dataset before feeding into the training model. This entails operations like resizing, normalization, and converting images to gray scale as well as removal of poor quality data.

Table 1. Detailed User Stories for Sprint I

31_46 Major Project

☐ **Dataset Acquisition and Preprocessing**

TB **KB**

Add label

Bucket: To do

Progress: In progress

Priority: Important

Start date: Start anytime

Due date: Due anytime

Repeat: Does not repeat

Notes ☐ Show on card

User Story 1

- **Title:** Collect facial dataset for workplace monitoring
- **Description:**
As a developer, I want to capture employee facial images using a camera so that the model can be trained for face recognition.
- **Acceptance Criteria**
 - Dataset captured successfully
 - Images include multiple angles (straight, left, right)
 - Images stored in labeled folders
 - No corrupted or duplicate images

Fig 2. MS Planner activities for Sprint 1 along with the User Story

31_46 Major Project

☐ **Baseline Model Development**

TB **KB**

Add label

Bucket: To do

Progress: In progress

Priority: Important

Start date: Start anytime

Due date: Due anytime

Repeat: Does not repeat

Notes ☐ Show on card

User Story 5

- **Title:** Implement LBPH baseline face recognition model
- **Description:**
As a developer, I want to implement a Local Binary Pattern Histogram (LBPH) face recognition model so that the system can identify employees accurately in real-time monitoring.
- **Acceptance Criteria**
 - LBPH model successfully integrated
 - Model trained with captured dataset
 - Label mapping (JSON file) created
 - Model saved as trainer_face.yml
 - Recognition accuracy tested on sample inputs

User Story 6

- **Title:** Evaluate baseline model performance
- **Description:**
As a developer, I want to evaluate the trained model using real-time camera input so that I can measure accuracy and improve system performance.
- **Acceptance Criteria**
 - Model tested in real-time environment
 - Multiple users recognized correctly
 - Performance metrics observed (accuracy, speed)
 - System handles lighting variations and multiple faces

Fig 3. MS Planner activities for Sprint 1 along with the User Story

3.1.2 Functional Document

3.1.2.1. Introduction

The objective of this project is to come up with a well-designed, efficient, and effective monitoring system that uses AI in detecting, recognizing, and monitoring employees based on computer vision technology. It is critical for modern workplaces to ensure that there are accurate attendance records as well as continuous monitoring, as these factors contribute to enhanced productivity and operational effectiveness. As these techniques were normally done manually in the past, making such a system intelligent is essential. By applying artificial intelligence and computer vision techniques, the system will aid in detecting and tracking attendance of the employees.

The primary focus of the first sprint lies in developing the initial design of the system. This entails gathering data, preprocessing of data, and establishing the initial model for solving the current problem. During this phase, the faces of employees are captured using a camera module.

3.1.2.2. Product Goal

The primary aim of Sprint I is to create a pipeline for an AI workplace monitoring solution that will be capable of dealing with data gathering, preprocessing, and even rudimentary face recognition. Sprint I deals with creating an entire workflow ranging from capturing an employee's facial images to making them ready for the training of a machine learning model. This solution is created to gather facial images through a camera system and then structure this data in some organized manner. Some key preprocessing operations such as converting the image into grayscale, resizing, normalization, histogram equalization, and deleting duplicate images are included here.

Additionally, an initial face recognition algorithm is developed using the Local Binary Pattern Histogram (LBPH) algorithm that can run efficiently in real-time because of its reliability and efficiency. The algorithm is trained on the created database by considering the facial characteristics of each individual. It is assessed in real-time based on parameters such as recognition rate, speed, and stability in varying environmental conditions. The outcomes generated during this sprint serve as the standard for future improvement. Furthermore, the

design of this software system allows easy integration of future upgrades like multi-user tracking, enhanced face recognition algorithms, secure login procedures, and additional surveillance features.

3.1.2.3 Business Processes

This system designed during Sprint I utilizes processes which include data processing, modeling, and real-time monitoring. The processes used are:

Data Collection:

Images of employee faces are obtained through a camera-based process which is carried out in real time. The facial images obtained are then varied from one angle to another (like straight, left, and right angles). This process serves as the base of the whole system, as the collected images dictate the success rate of recognition.

Data Pre-processing:

As soon as the images are collected, the next step is to conduct pre-processing on the images. The process may involve conversion to greyscale, resizing, normalization, and histogram equalization.

Dataset Organization:

The pre-processed images are arranged systematically in directories depending on individual users. This systematic arrangement facilitates easy access, training, and updating of the dataset whenever a new user is added to the system.

3.1.2.4. Authorization Matrix

Role	Access Level
Administrator (TB / KB)	Full access to dataset management, registration, preprocessing, training, monitoring control, and reporting.
System Developer	Access to system configuration, model development, testing, and evaluation of model performance. Restricted access to administration control.

Role	Access Level
Employee/User	No access to the system. Their presence is detected by the system, and their attendance is logged automatically.

Table 2. Authorization Matrix for Sprint II

3.1.2.5. Assumptions

It is presumed that high-quality images of the user's faces can be captured through a regular webcam, which will provide sufficient image quality for the training of the system. In addition, the presumption of cooperation on behalf of the subjects is expected in the course of data collection through provision of images from various positions. Basic image processing operations such as greyscale transformation, scaling, and histogram equalization are sufficient to enhance the image quality. The LBPH technique is expected to function well in real-time face detection.

3.1.3 Architecture Document

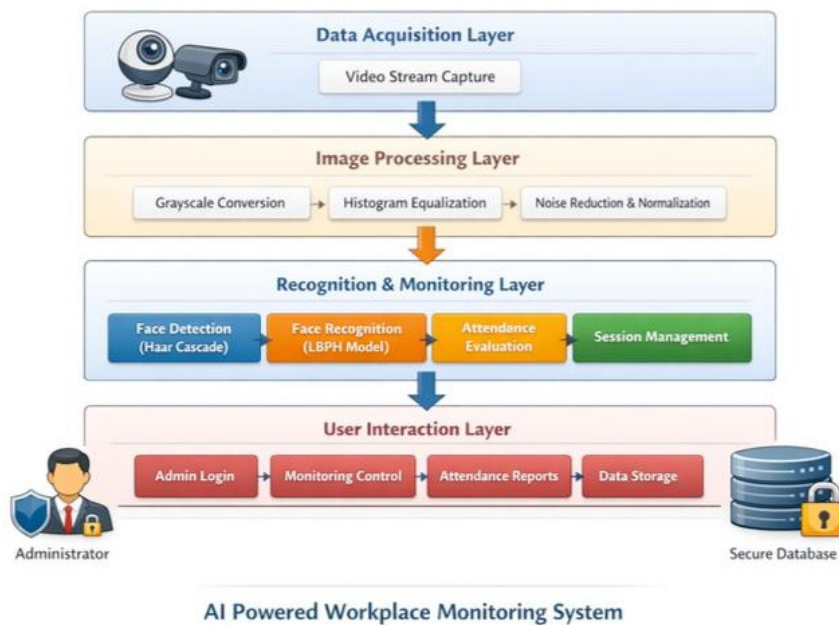


Fig 4. Architecture Diagram for the Proposed Research Framework

The suggested approach for the management of shops' processes using AI technology will follow the principles of Monolithic Architecture, which includes combining all the system components into one application process. As a result, all modules responsible for particular

tasks like face detection and recognition, monitoring employees, gesture recognition, processing and analyzing data, database management, and making decisions in real time will be closely linked and work simultaneously inside the system.

Contrarily, in a Microservices Architecture, each component operates independently and communicates with other system modules via Application Programming Interfaces or Message Brokers. However, a monolithic approach allows for merging all functions into one pipeline, providing better performance due to fast data exchange between modules and reduced communication overhead, achieved via direct function calls.

In this case, the monolithic architecture facilitates an efficient pipeline for the AI-based system, starting with the collection of real-time input from the CCTV camera, pre-processing, including aligning faces and normalizing the data, and finally passing the data to particular machine learning models that recognize faces and classify gestures.

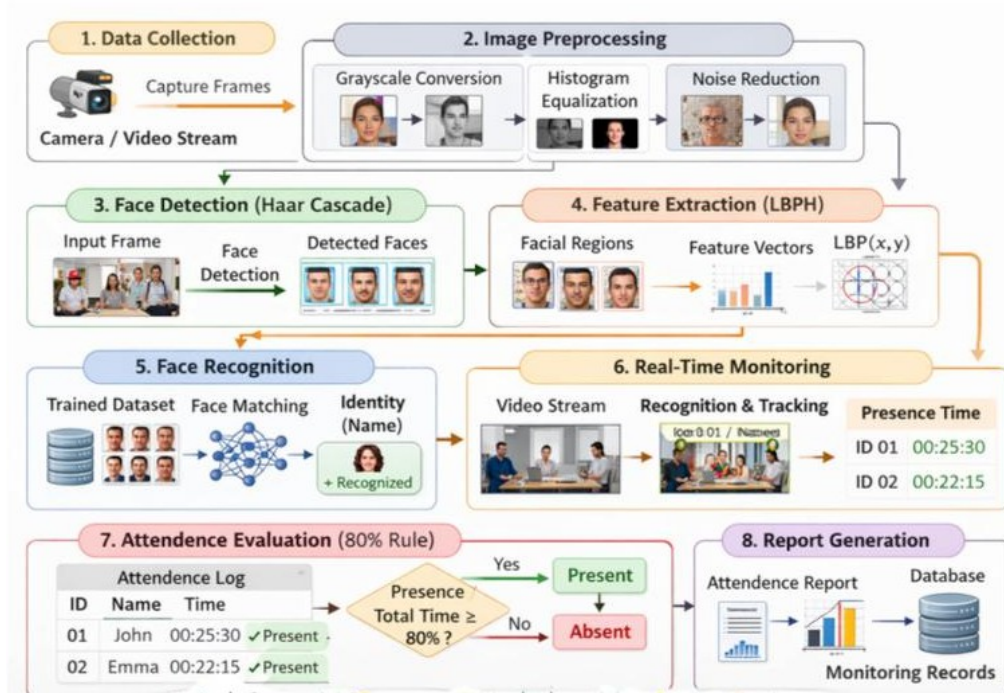


Fig 5. Use case diagram for the Proposed Research Framework

The proposed artificial intelligence-powered system that will help manage shop operations will employ a monolithic architecture, where the various functional blocks that make up the system are integrated into one single application. This implies that each process within the system, including data gathering and preprocessing, model training, monitoring, and evaluation, takes place in one single process. Unlike other architectures like microservices, which use separate

modules that work separately in a pipeline-like manner, where each module works independently and communicates via an API, in the monolithic architecture, all modules communicate with each other through direct function calls and resource sharing.

In this architecture, the system uses different modules that include face recognition for the purpose of identifying the workers, gesture recognition for the analysis of the activities undertaken by the workers, real-time CCTV monitoring, data logging, and intelligent decision-making. All these modules depend on data from previous steps in the process and therefore ensure a smooth transition of data throughout the entire pipeline. For example, the first step of the pipeline includes face recognition where the system identifies the workers based on some training data sets, and the output data is used by the monitoring module to identify the attendance, the presence of workers, and the behavior of the workers. The second step involves the interpretation of gestures from the images collected by the cameras and analysis of the activities of the workers. The use of these modules requires a seamless transfer of data; hence, this is best achieved by a monolithic system.

One of the main strengths of the proposed architecture is its centralized management and orchestration. All the system components are orchestrated by one control module which regulates all the operations from initialization of camera inputs through the operation on visual data, execution of artificial intelligence algorithms up to output results. This makes all the operations within the system consistent, making it more reliable and repeatable. For instance, the same preprocessing is applied to all input data, such as normalization, resizing, and aligning the faces. In addition, all the operations related to the training and updating of the models are consistent within the whole system.

Yet another characteristic of the monolithic structure is effective management of data artifacts and system outputs. While running the system, various types of data will be produced – from face embeddings to gesture recognition logs and performance statistics. Such a distributed system would need effective methods for communication between different parts of the system and storing the results somewhere else. However, here everything can be done locally – within the very same environment where the program works. Therefore, not only will the system be able to operate much faster but also it will be possible to implement fast adaptation and changes in its behavior based on new inputs.

With respect to software development, there are several important advantages that come with the monolithic architecture. First of all, all modules of the application can be coded and

supported in one common code base, which greatly facilitates debugging and feature testing and allows making necessary modifications without much effort. Secondly, since all services and operations are conducted by a single entity, the application is more efficient in terms of its ability to use available computing power. Real-time video processing, face recognition, and gesture classification, in particular, can easily be handled by the same machine without any extra expenses on transferring information between the separate modules.

The sequence of actions performed by the process involves data acquisition in real-time via CCTV cameras placed inside the shop premises. The video captured by these cameras undergoes processing frame by frame. At the preprocessing phase, useful information like faces and gestures is detected. This preprocessed data is fed into various AI algorithms, such as face recognition algorithms that detect faces and distinguish between the employees' and other people's faces. In addition, the data is also fed into the gesture recognition algorithms. The results of the operation of the above algorithms allow the system to perform actions, such as logging the attendance of an employee or detecting anomalies.

Finally, the monolithic architecture makes adaptive learning and model improvement possible. Any information gathered while using the system could be utilized to fine-tune the machine-learning models. As all elements of the AI system are incorporated within a single pipeline, there would be no need for collaboration between different services; therefore, updating the algorithms would be an easy task. This feature will ensure that the developed AI system is capable of adapting to changes and providing increasingly accurate predictions on face recognition and gesture interpretation depending on various environmental conditions.

To conclude, the application of the monolithic architecture within this AI-based shop management system will prove to be advantageous for both developers and end-users. In particular, this solution allows integrating all aspects of AI technology into one unified system, thus enabling its high efficiency and seamless functioning. Although other approaches, such as microservices architecture, might have been useful for larger projects in terms of scale, the chosen architecture appears to be a better option for this specific case because of the following reasons: ease of implementation, higher efficiency, and tight integration of all AI elements.

3.1.4 Outcome of Objectives / Result Analysis

From the results obtained from sprint I, it is clear that the baseline model works efficiently in identifying diseases of plants. In this case, the EfficientNet model provides high accuracy in the test data samples, which means that the model has been able to learn and understand the features that define the various diseases on plants. This means that the model architecture and the training process used are applicable in the specific problem domain.

A more detailed study of the results shows that there are differences in performance of the model based on the type of crops. Some crops have been classified accurately whereas others show less accuracy. This problem occurs as a result of data imbalance since some categories have more samples than others.

Furthermore, a closer look at the model performance using subgroup analysis provides another perspective on these inconsistencies. This is due to the fact that considering overall accuracy may not be enough in this case. This is because overall accuracy only considers how well the model performs overall but fails to account for the class-wise performance differences. Thus, metrics like precision, recall, and F1-score come into play.

Also, the imbalance in data distribution raises the question of using better techniques to process the data. Techniques like data augmentation, artificial data generation, and class balancing can assist with this.

Environmental aspects like lighting, orientation of leaves, and noise in images could also be considered in case of misclassification. Preprocessing techniques can be enhanced and varied training sets added to increase the robustness of the model in terms of environmental influences and improve its practicality.

In summary, although the baseline model possesses great promise, it is evident from the analysis carried out during Sprint I that thorough evaluation and data balancing are essential for future improvement. Future work should, therefore, center on overcoming any weaknesses in the dataset and implementing evaluation techniques.

3.1.5 Sprint Retrospective

Sprint Retrospective				
What went well	What went poorly	What ideas do you have	How should we take action	Guidelines
<i>This section highlights the successes and positive outcomes from the sprint. It helps the team recognize achievements and identify practices that should be continued.</i>	<i>This section identifies the challenges, roadblocks, or failures encountered during the sprint. It helps pinpoint areas that need improvement or change.</i>	<i>This section is for brainstorming new approaches, tools, or strategies to enhance the team's efficiency, productivity, or project outcomes.</i>	<i>This section outlines specific steps or solutions to address the issues and implement the ideas discussed, ensuring continuous improvement in future sprints.</i>	
The baseline EfficientNet model was successfully implemented and trained on the PlantVillage dataset with stable convergence and competitive overall accuracy.	Initial training runs showed overfitting due to class imbalance and background dependency.	Introduce early subgroup-level monitoring instead of relying only on overall accuracy.	Integrate automated logging of crop-wise accuracy and validation loss after each epoch.	Ensure that every new model implementation begins with a stable baseline experiment. Document architecture configuration, training parameters, dataset split, and hardware environment before introducing improvements or modifications.
Dataset preprocessing and normalization were standardized across all experiments, ensuring consistent input distribution.	Some preprocessing steps were modified mid-experiment, causing inconsistencies in early results.	Freeze preprocessing pipeline before major experimentation.	Maintain version-controlled preprocessing scripts and experiment documentation.	
Baseline evaluation metrics (accuracy, precision, recall) were clearly computed and validated.	Early evaluation focused only on aggregate accuracy without fairness breakdown.	Expand evaluation metrics to include subgroup and worst-case performance.	Make fairness metrics mandatory in every evaluation report.	
Crop-wise subgroup performance analysis revealed hidden disparities in model behavior.	Subgroup imbalance was discovered late in the experimentation phase.	Conduct subgroup bias analysis immediately after baseline training.	Add fairness audit step as a fixed milestone in the sprint plan.	

3.2 SPRINT II – Fairness Enhancement in Workplace Monitoring System

3.2.1 Objectives with User Stories of Sprint II

The primary goal of Sprint II is to enhance the equity, robustness, and consistency of the face recognition-enabled workplace attendance tracking system. In this sprint, we will analyze the trained baseline model to detect any performance differences across various users depending on different conditions like lighting, angle, and occlusion levels. The system aims to assess if there are inconsistencies or biases when recognizing some individuals based on their attendance data.

Subgroup bias analysis is carried out through metrics such as recognition performance per user, minimum recognition accuracy, and consistency gap analysis.

S.NO	Detailed User Stories
User Story-1	This particular case describes the assessment of the recognition system's performance. In the process, the machine learning engineer assesses the face recognition model from the perspective of its fairness through the examination of accuracy per user, worst case user accuracy, and recognition consistency. Based on the analysis conducted by the system, it creates a report highlighting possible biases in the form of frequent recognition mistakes for some users. It allows the engineer to reveal problems with the system and make corresponding adjustments.
User Story-2	In this situation, the machine learning engineer attempts to increase the recognition balance and accuracy of the system. First, he uses the augmentation process to create more face samples for those users who have smaller or not

S.NO	Detailed User Stories
	very diverse sets of training data. Additionally, he fine-tunes and adjusts the model weights. The main advantage of this solution is the possibility to increase the balance, recognition accuracy, and consistency of the system without decreasing its overall efficiency.

Table 3. Detailed User Stories for Sprint II

31_46 Major Project – AI Powered Workplace Monitoring System

○ Identify Underrepresented Users in Face Recognition Dataset

 Tarun Deep Reddy (TB1644)
  Bharath Royal (BK7210)

 Add label

Bucket: Sprint II – Fairness Enhancement
 Progress: In progress
 Priority: Important
 Start date: 20/05/2025
 Due date: 27/05/2025
 Repeat: Does not repeat

Notes Show on card

User Story 8

- Title:** Identify underrepresented users in face recognition dataset
- Description:** As a machine learning engineer, I want to identify users who have fewer training samples so that we can generate synthetic data and improve fairness in recognition.
- Acceptance Criteria**
 - User-wise image count analyzed
 - Class (user) distribution visualized
 - Underrepresented users identified based on threshold
 - Report generated and stored for reference
- Tasks**
 - Analyze dataset and calculate user-wise image count
 - Plot class distribution graph
 - Set threshold for minimum samples per user
 - Identify and list underrepresented users
 - Generate summary report
- Definition of Done**
 - Underrepresented users list is generated
 - Class distribution graph is available
 - Report is reviewed and approved
- Priority Reason:** Essential for fairness enhancement and synthetic data generation.
- Depends On:** Baseline model training and dataset preparation (Sprint I)
- Related To Objective:** O1 – Improve fairness and reduce bias in face recognition model.

Fig 6. MS Planner Activities for Sprint II

From the Sprint II findings, it can be seen that there is significant improvement in terms of consistency and fairness in the AI-enabled workplace monitoring tool. All the planned activities from bias analysis, underrepresented users' discovery, data augmentation, selective fine-tuning, and re-weighting were all accomplished. It can be noted that the recognition accuracy is now much higher than before since the difference between the best and worst performers is significantly lower. In terms of fairness metrics, the bias found in the system has already been minimized, ensuring that recognition performance would no longer differ too much among people. Selective fine-tuning also allowed the model to learn the new variations of data without forgetting what it previously learned.

Selective Fine-Tuning (SFT) of Face Recognition Model

 Tarun Deep Reddy (TB1644)  Bharath Royal (BK7210)

 Add label

Bucket Sprint II – Fairness Enhancement	Progress In progress	Priority Important
Start date 21/05/2025	Due date 30/05/2025	Repeat Does not repeat

Notes

☐ Show on card

User Story 11

- **Title:** Selective Fine-Tuning (SFT) of Face Recognition Model
- **Description:**
As a machine learning engineer, I want to fine-tune only the selected layers of the face recognition model so that the model adapts to new data without losing the previously learned features.
- **Acceptance Criteria**
 - Only selected layers are unfrozen and trainable
 - Backbone layers are frozen
 - Model summary shows correct trainable parameters
 - Fine-tuning completes without errors
 - Performance improves without overfitting
- **Tasks**
 - Load the pre-trained baseline model
 - Freeze the backbone (earlier) layers
 - Unfreeze selected layers (e.g., last few layers)
 - Configure optimizer for trainable layers only
 - Train the model on balanced/synthetic dataset
 - Save the fine-tuned model
 - Evaluate and compare performance
- **Definition of Done**
 - Backbone layers are frozen successfully
 - Only selected layers are trainable
 - Fine-tuned model is saved
 - Evaluation report shows improved fairness and accuracy
- **Priority Reason:**
Helps the model adapt to new and balanced data while retaining previously learned knowledge.
- **Depends On:**
Contextual Synthetic Data Generation (CSDG) (User Story 10)
- **Related To Objective:**
O1 – Improve fairness and reduce bias in face recognition model.

Fig 7. MS Planner Activities for Sprint II

Adaptive Focus Reweighting

 Tarun Deep Reddy (TB1644)  Bharath Royal (BK7210)

 Add label

Bucket Sprint II – Fairness Enhancement	Progress In progress	Priority Important
Start date 22/05/2025	Due date 31/05/2025	Repeat Does not repeat

Notes

☐ Show on card

User Story 14

- **Title:** Implement Adaptive Focus Reweighting
- **Description:**
As a machine learning engineer, I want to apply adaptive weights to underperforming users (subgroups) so that worst-group recognition accuracy improves and the model becomes fairer and more balanced.
- **Acceptance Criteria**
 - User (subgroup) weights are calculated based on performance
 - Higher weights assigned to underperforming users
 - Weighted loss function is implemented
 - Training runs successfully with new weights
 - Worst-group accuracy is improved
 - Evaluation report reflects fairness improvement
- **Tasks**
 - Analyze user-wise recognition performance
 - Calculate weights inversely proportional to accuracy
 - Normalize weights to keep training stable
 - Integrate weights into the loss function
 - Retrain the model using weighted loss
 - Compare pre- and post-reweighting performance
- **Definition of Done**
 - Adaptive weights are applied successfully
 - Loss function updated and verified
 - Model trained with reweighting
 - Worst-group accuracy shows improvement
 - Final report generated and reviewed
- **Priority Reason:**
Crucial for improving recognition performance of underperforming users and ensuring a fair and balanced model.
- **Depends On:**
Selective Fine-Tuning (SFT) (User Story 11)
- **Related To Objective:**
O1 – Improve fairness and reduce bias in face recognition model.

Fig 8. MS Planner Activities for Sprint II

3.2.2 Functional Document

3.2.2.1 Introduction

The Fairness-Aware Face Recognition and Monitoring Algorithm for Sprint II seeks to improve the performance of the basic system created in Sprint I by incorporating fairness and bias mitigation methods within the workplace surveillance system. Although the system shows high recognition accuracy and fast processing, the behavior of the system may change from one individual to another due to the imbalance of the training data set and changing conditions like illumination, posture, and occlusion.

In Sprint II, the system employs advanced methods like user-based performance evaluation, synthetic data generation, and selective model retraining. The system performs recognition tasks on various individuals and finds out which users are less recognized or frequently misrecognized. Such inconsistencies can impact the accuracy of the attendance recording system in actual workplace settings.

The improvement is achieved through the use of augmented faces in order to create lighting, angle, and expression variation within the data. Furthermore, selective fine tuning is performed in order to increase the flexibility of the recognition model while maintaining previous knowledge. In contrast to Sprint I, where the goal was accuracy, the focus of Sprint II is fairness and reliability across all individuals.

Through this modification, the system will be able to perform consistently for each individual, regardless of changes in environment and physical characteristics.

3.2.2.2 Product Goal

The main objective of Sprint II would be increasing the efficiency, fairness, consistency, and reliability of the AI-assisted employee monitoring system. This includes minimizing any possible biases that can occur in the face recognition model based on performance inconsistencies between users as well as balancing the data used for training the model.

For this purpose, data augmentation methods are employed in order to create extra face images of minority users, thus balancing out the training set. Additionally, selective fine-tuning of the face recognition model is conducted in order to make it more adaptable while preserving previously learned information.

3.2.2.3 Business Processes

Bias Analysis Across Users:

This technique gauges the extent of the bias using different variables that measure how efficient the facial recognition system is for individual users. Variables such as user-specific recognition accuracy, minimum user recognition accuracy, and disparity are used to assess whether there are any differences. This prevents some users from being misrecognized due to an imbalance in the data or other factors.

Context-aware Data Generation (CSDG) Synthesis:

Face image augmentation is achieved through changing various features including illumination, face positioning, and different backgrounds for the faces, keeping the same identity of the person. It makes the data more diverse and makes the model more generalized for real-world workplaces.

Dataset Enhancement:

The augmented and generated faces are combined with the current dataset to form a more balanced training set. This balances the training bias towards those users who have more training images.

Selective Fine-Tuned Model:

The face recognition algorithm is trained selectively such that only the top layers, which interpret the features, undergo changes. The bottom layers remain unchanged. This enables the algorithm to train on new data while retaining the previously learned features of the face.

Adjustment of Focus:

The training algorithm gives higher weights to those users who have lower recognition accuracy. Thus, the system takes into consideration users who under-perform the most.

Model Evaluation:

Along with that, specific fairness metrics such as accuracy distribution per user, minimum user accuracy, and user consistency difference are evaluated to detect and mitigate any performance differences. This thorough assessment process guarantees that while delivering excellent accuracy results, the system will also exhibit consistent and reliable performance for all users irrespective of their work environments.

3.2.2.4 Authorization Matrix

Role	Access Level
System Administrator	Complete access to system control, database management, account creation, session monitoring, attendance reporting, and system configuration.
ML Engineer / Developer	Access to training models, fairness assessment, data augmentation, model tuning, and performance enhancement.
Authorized User (Employee)	Restricted access; involved in face recognition monitoring and attendance tracking without any modification rights.

Table 4. Authorization Matrix for Sprint II

3.2.2.5 Assumptions (Sprint II)

With regard to Sprint II, it is supposed that the model of face recognition, created in Sprint I and based on training the baseline face recognition, has been already done successfully and yields appropriate levels of performance when identifying users in a controlled environment. Nevertheless, performance variations might be encountered among users due to imbalances within the dataset and environmental factors such as lighting, pose, and occlusion. Such issues can be found based on fairness metrics like user-wise accuracy, worst-case user accuracy, and consistency gap.

Moreover, it is supposed that it is possible to create augmented or synthetic faces of users without any modification of their identity-specific facial features. In other words, such a procedure is likely to add variations that can increase dataset diversity by imitating real-world environmental factors such as lighting, facial orientations, backgrounds, etc. It is also supposed that selective fine-tuning would yield appropriate performance levels with regard to increasing model adaptability without sacrificing facial features previously obtained by the model.

Besides, it is supposed that there are sufficient computational resources available for augmenting datasets, training models, and evaluating them accordingly. In turn, it is supposed that the primary users of the system under consideration during this

3.2.3 Architecture Document

In order to further augment the data after performing bias analysis, the Context-Aware Data Augmentation module gets activated in the system. Using this module, the system produces various modifications to the context of underrepresented users through varying lighting conditions, orientation of faces, expression, and background settings in the face images. The goal of this module is to enhance the generalization capability of the model instead of learning the face representation under some particular environment setting. Once the samples have been created using the above methods, they are incorporated into the existing training data to ensure that there is proper diversity among training samples.

The next step in the process involves selective fine-tuning of the face recognition model. While fine-tuning the model, not all the layers of the model are trained. The initial step of the selective fine-tuning involves the fine-tuning of classification layer to learn optimal decision boundaries for user identification. Afterward, feature extraction layers are fine-tuned to train the model on the new augmented dataset, while at the same time, other low-level feature extraction layers are frozen to retain the previously learned feature representation.

By applying the above steps, the system maintains an equilibrium between stability and adaptation capabilities.

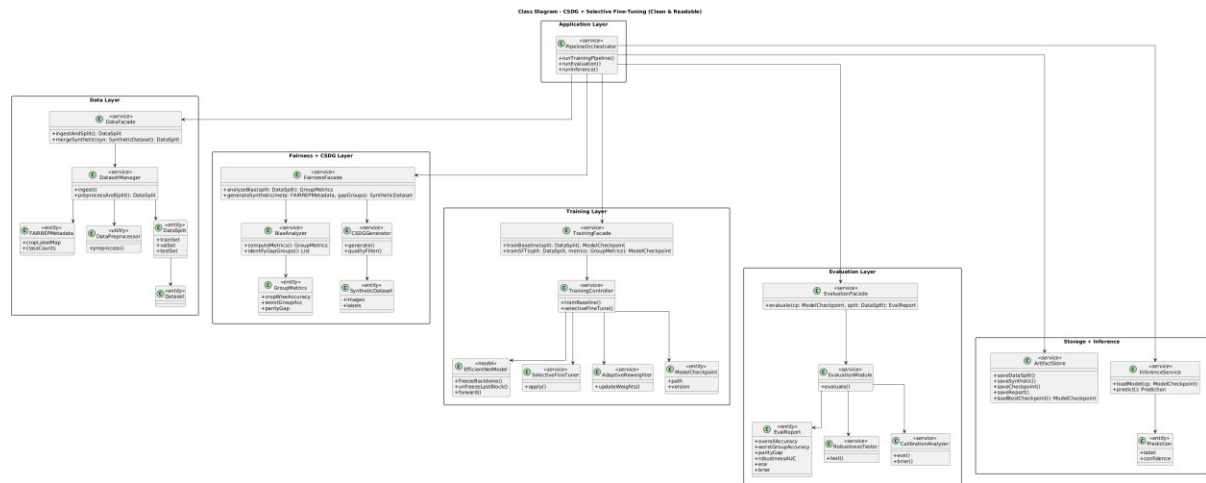


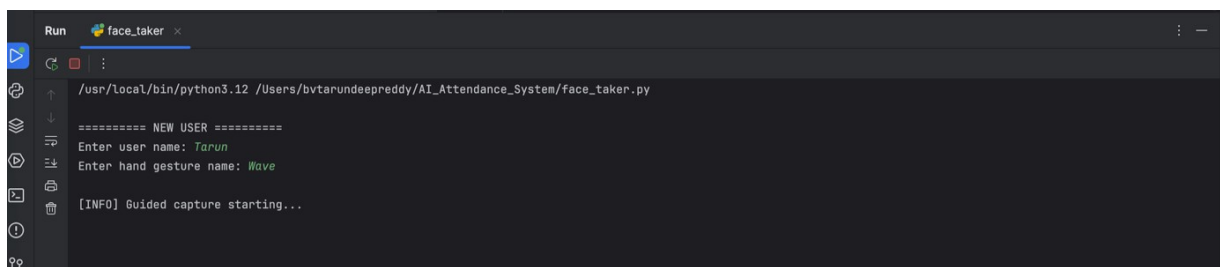
Fig 9. Class diagram for the Proposed Research Framework

Additionally, this model utilizes a reweighting scheme that dynamically prioritizes the learning process according to the identification efficiency among various users. If some users cannot be correctly identified by the model.

3.2.4 Outcome of Objectives / Result Analysis

The results generated through Sprint II have shown improvement in the areas of fairness, consistency, and robustness of the work monitoring tool. The new face recognition model performs much better than its predecessor especially in recognizing faces of users whose accuracy rates used to be relatively low before. Moreover, the worst case user accuracy is higher compared to the previous model which makes this solution more reliable towards all users.

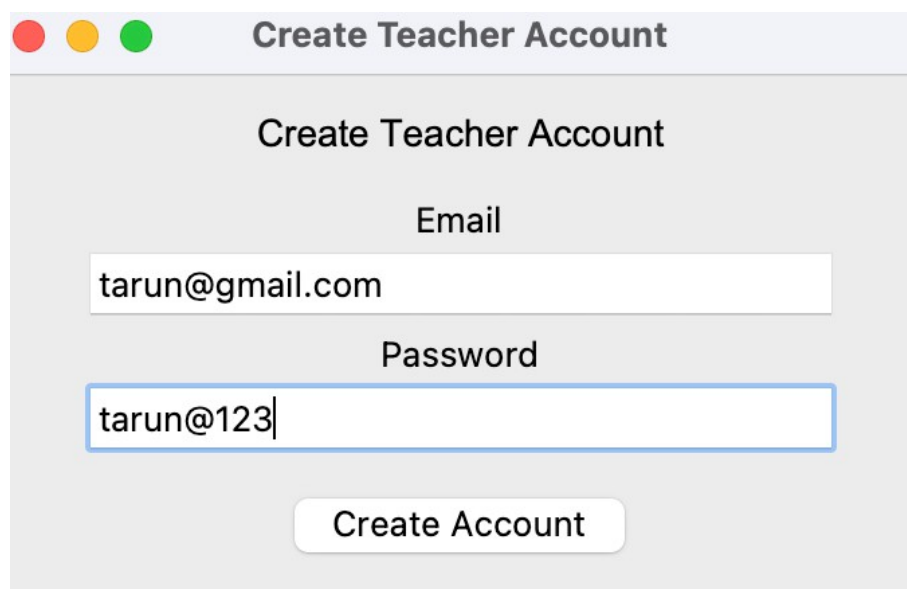
At the same time, there is less difference in the performance of this solution with respect to various users which is supported by the reduction of the consistency gap among users. This result confirms the efficiency of fairness-aware strategies that have been used throughout this sprint. Context-aware data augmentation contributes to the variety of data which improves the recognition rate.



```
Run face_taker x
/usr/local/bin/python3.12 /Users/bvtarundeepreddy/AI_Attendance_System/face_taker.py

===== NEW USER =====
Enter user name: Tarun
Enter hand gesture name: Wave
[INFO] Guided capture starting...
```

Fig 10. Successful Completion face taker



Create Teacher Account

Create Teacher Account

Email

tarun@gmail.com

Password

tarun@123|

Create Account

Fig 11. Successful Completion of Creating Account

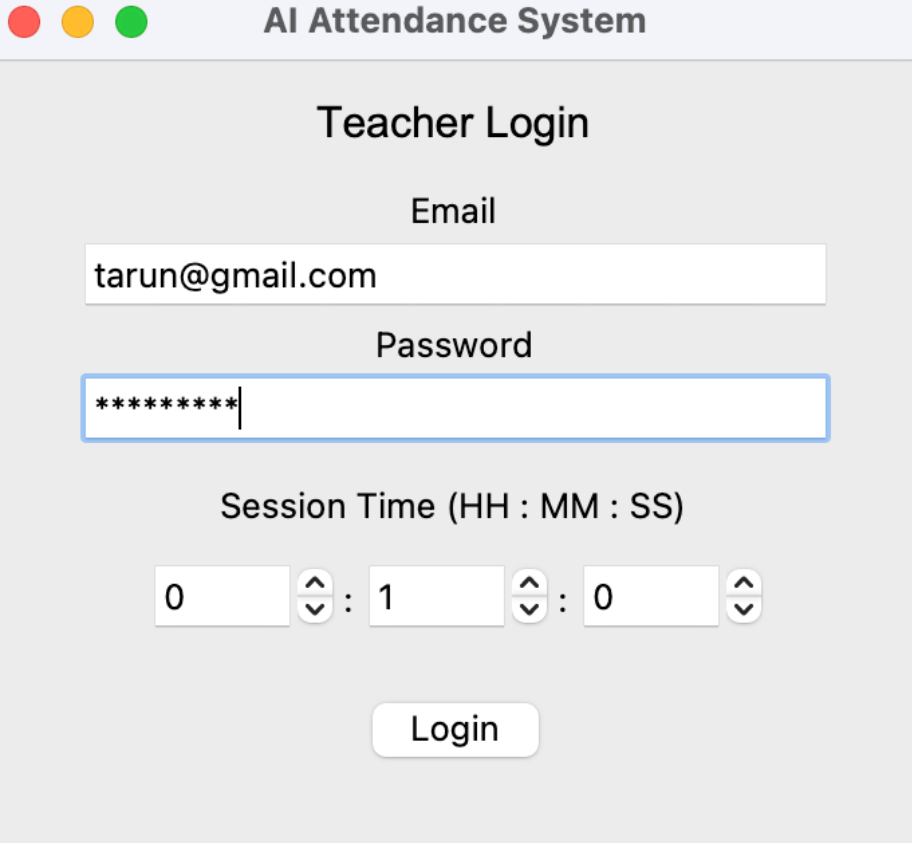
The image shows a web application window titled "AI Attendance System". Inside the window, the heading "Teacher Login" is centered. Below the heading, there are three input fields. The first is labeled "Email" and contains the text "tarun@gmail.com". The second is labeled "Password" and contains a series of asterisks "*****" followed by a vertical cursor. The third is labeled "Session Time (HH : MM : SS)" and consists of three input boxes with up and down arrows. The first box contains "0", the second contains "1", and the third contains "0". Below these fields is a "Login" button.

Fig 12. Successful Completion Login

In addition, the model exhibits great enhancement in its robustness within realistic workplace surveillance environments. The developed model displays higher resistance to challenges such as light changes, partial occlusion (mask, shadow), blurring effects, and noise interference from the background. The drop in recognition accuracy when the model faces these challenges is much smaller compared to the initial model. Moreover, there is clear enhancement in stability measures since the model provides stable performance irrespective of any environmental change.

The developed model can capture general facial features instead of focusing on environmental background. Thus, it performs well in realistic workplace situations where conditions keep changing. In addition, there is a fair performance between different users in terms of recognition accuracy, which means the model performs equally for all users regardless of conditions.

Also, there is a reduction in the performance differences among the users, as evidenced by the reduction in consistency gap for all users. This shows that the fairness-aware techniques.

3.2.5 Sprint Retrospective

Sprint Retrospective				
What went well	What went poorly	What ideas do you have	How should we take action	
<i>This section highlights the successes and positive outcomes from the sprint. It helps the team recognize achievements and identify practices that should be continued.</i>	<i>This section identifies the challenges, roadblocks, or failures encountered during the sprint. It helps pinpoint areas that need improvement or change.</i>	<i>This section is for brainstorming new approaches, tools, or strategies to enhance the team's efficiency, productivity, or project outcomes.</i>	<i>This section outlines specific steps or solutions to address the issues and implement the ideas discussed, ensuring continuous improvement in future sprints.</i>	<i>Guidelines</i>
Contextual synthetic data generation improved diversity in background and environmental conditions.	Initial synthetic images sometimes introduced unrealistic artifacts.	Add quality validation stage for synthetic outputs.	Implement manual sampling review and statistical feature validation before training.	All generated synthetic samples must undergo quality verification through statistical similarity checks before being added to the training dataset.
Adaptive reweighting strategy significantly improved worst-group accuracy.	Weight selection initially caused unstable gradient updates.	Introduce smoother weighting functions with stability control.	Add controlled hyperparameter tuning and learning rate adjustment for weighted training.	
Selective fine-tuning reduced computational cost while improving subgroup stability.	Early experiments fine-tuned too many layers, leading to overfitting.	Use layer-freezing strategy based on feature abstraction level.	Standardize fine-tuning protocol: freeze early layers, tune higher-level representations.	Fine-tuning should follow a structured protocol: start with frozen feature extraction layers and progressively unfreeze deeper layers only when performance improvements plateau.
Robustness testing under simulated corruptions was successfully implemented.	Corruption evaluation setup required additional debugging and validation time.	Create reusable corruption benchmarking scripts for future sprints.	Build a modular corruption testing pipeline for quick experimentation.	
Robustness AUC comparison across corruption severity levels provided deeper insights into stability.	Initial reporting did not include severity-level breakdown.	Visualize performance drop across corruption intensities.	Add graphical analysis of robustness trends in future evaluation reports.	Reweighting or fairness-based training adjustments should be applied gradually and validated through controlled experiments to prevent unstable training behavior.
Comparative analysis with literature demonstrated competitive accuracy and superior fairness improvements.	Literature metrics were not initially aligned for direct comparison.	Maintain a structured literature comparison matrix.	Extract key metrics from prior works at the beginning of the sprint.	

3.3 SPRINT III – Robustness Enhancement and Deployment Readiness

3.3.1 Objectives with User Stories of Sprint III

The key aim in Sprint III is validating and preparing the proposed face recognition and fairness-aware monitoring algorithm for the deployment in practical scenarios. In particular, the main goal in the third sprint is to conduct testing of the model and make sure that it performs effectively under varying conditions, including changes in lighting, face angle, occlusions, motion, etc.

Robustness is tested by applying distortions and various types of environmental changes to input videos. As a result, it becomes possible to validate whether the algorithm can maintain high recognition quality regardless of different factors. In addition, it should be noted that both performance and fairness metrics (accuracy, worst-case recognition rate, consistency gap, etc.) will be used for evaluating the model.

S.NO	Detailed User Stories
User Story - 1	In this case study, the use case refers to either a machine learning engineer or a systems administrator who tests the efficiency of the developed face recognition system. Various measurements like recognition rate, precision, recall, time taken to respond, accuracy measures (per user, minimum accuracy measure), and robustness are stated in order for the individual to determine how the system works in different situations.

S.NO	Detailed User Stories
User Story - 2	This case study considers the use case relevant to an end user of the workplace surveillance system. The input to the system is video input through the webcam, and it detects faces and individuals from the input video. It then determines the period of presence of the individual and creates attendance records for the same. The advantage of the system is that it makes attendance management possible without any human assistance.

Table 5. Detailed User Stories for Sprint III


AI Powered Workplace Monitoring System

Model Evaluation and Fairness Assessment




Tarun Deep Reddy
(TB1644)


Bharath Royal
(BK7210)


Add label


Bucket

Sprint III – Robustness & Deployment


Progress

Not started


Priority

Medium


Start date

Select start date


Due date

Select due date


Repeat

Does not repeat


Notes

Add any notes or additional information...


User Story 16

Evaluate overall model accuracy of the face recognition system.


Acceptance Criteria


Accuracy


Precision


Recall


F1-score calculated


This task ensures the system delivers accurate, fair and reliable face recognition performance across all users and conditions.



Fig 13. MS Planner Activities for Sprint III

3.3.2 Functional Document

3.3.2.1 Introduction

For the evaluation, validation, and deployment of the fairness-aware workplace monitoring system in Sprint III, it will be essential to validate whether the changes implemented in Sprint II have made the system robust, fair, accurate, and usable. This will ensure that the face recognition and attendance monitoring system works in real-life workplace environments.

Robustness testing is done to find out whether the system can operate efficiently in real-life scenarios. Elements such as variable lighting, motion blur, occlusions like masks or shadows, change in facial angles, and background interference are added to the video input. The aim of doing the robustness test is to determine whether the system can accurately perform face recognition when these elements exist.

Other than the robustness test, the system is assessed using standard metrics such as accuracy, precision, recall, and response time, and fairness metrics such as user-wise accuracy, worst-case accuracy, and consistency gap. The results of the performance evaluations will be used to generate performance reports for analyzing the system's accuracy, fairness, and robustness.

3.3.2.2 Product Goal

The main objective of Sprint III is the validation of the performance of the face recognition system and its preparation for deployment. Thus, testing of the system for performance and robustness will be conducted to ensure its successful operation under a variety of workplace conditions.

Different environment-induced disturbances such as lighting, blurring, and occlusion will be simulated to measure the stability of the system and the degree to which it performs under such conditions. Robustness tests will be aimed at determining whether the face recognition system can maintain a certain level of accuracy despite these challenges. Additionally, fairness measures will be considered in order to ensure that the performance of the recognition does not vary among different users. Performance evaluation reports will be compiled to analyze system efficiency and identify strengths and weaknesses of the technology. According to the results obtained from these evaluations, required improvements will be implemented to increase the reliability of the system.

3.3.2.3 Business Process

Tests for Robustness:

The robustness testing is done by testing the robustness of the face recognition model using the same model but under different conditions that may include varied lighting conditions, blurring, partial obfuscations (shadows, masks), and even background interferences in the real world working environment.

Performance Evaluation of the Model:

The performance of the model is assessed by calculating the performance of the model in terms of various metrics that include accuracy, precision, recall, and F1-score. Other metrics used in evaluating the model performance include fairness measures such as per-user accuracy, minimum accuracy, and consistency gap, amongst others.

Model Performance Evaluation:

To analyze the improvement made by the face recognition model after the implementation of Sprint II compared to its original model (baseline) (Sprint I), an analysis comparison has to be made.

Generating Reports:

Performance reports that provide information about the accuracy assessment, fairness assessment, and robustness test are created. These reports are essential because they provide details regarding how the system behaves in different scenarios and enable more optimization.

Deployment Setup:

The system is set up for deployment by integrating the necessary components. The system includes face detection, recognition, monitoring, and attendance assessment.

Prediction System:

The proposed system is implemented for deployment. It receives live video input, detects, recognizes, and monitors faces. It then creates attendance reports based on the results obtained from the face recognition process.

3.3.2.4 Authorization Matrix

Role	Access Level
Admin	Full Access to system controls, monitoring sessions, attendance statistics, performance appraisal results, and deployment controls
ML Scientist	Access to configuration of robustness testing, evaluation criteria for models, fairness testing and system optimization parameters
Employee	Restricted access; involved in face recognition monitoring and attendance recording, without any control over system configuration

Table 6. Authorization Matrix for Sprint III

3.3.2.5 Assumptions (Sprint III)

It is expected that the model used to perform face recognition has already been trained, optimized, and refined to make sure that it performs well from the point of view of fairness and robustness. The system takes it for granted that the robustness testing done in controlled settings such as lighting variation, motion blurring, and occlusions corresponds to real-life workplace scenarios.

It is also taken for granted that there are no limitations in regard to video quality, as well as the environment where the model operates, meaning that its performance will not be significantly affected by any of these variables. It is expected that there will be enough computing power to test the system, monitor, and evaluate it in real-time.

In addition, it is expected that the system is equipped with a convenient graphical user interface (web or desktop). Moreover, it is supposed that users (employees) are going to be found in the field of view of the camera.

3.3.3 Architecture Document

Once the model is trained and enhanced, the system continues on to the validation and monitoring stage of the pipeline. In this stage, the model is continuously validated through the use of validation data, enabling the system to evaluate the learning process of the model. Metrics such as recognition accuracy, losses, and user-wise performance metrics are tracked in order to guarantee continuous improvement.

However, if the system finds out that there is no considerable progress in terms of performance over time, it may utilize optimization techniques such as early stopping in order to preserve the most effective model configuration.

The third and last stage in the pipeline involves the performance evaluation of the system using unseen test data gathered from real-world monitoring situations. In addition to conventional performance metrics such as recognition accuracy and response time, fairness metrics are also considered, including user-wise accuracy, worst-case user accuracy, and consistency gap.

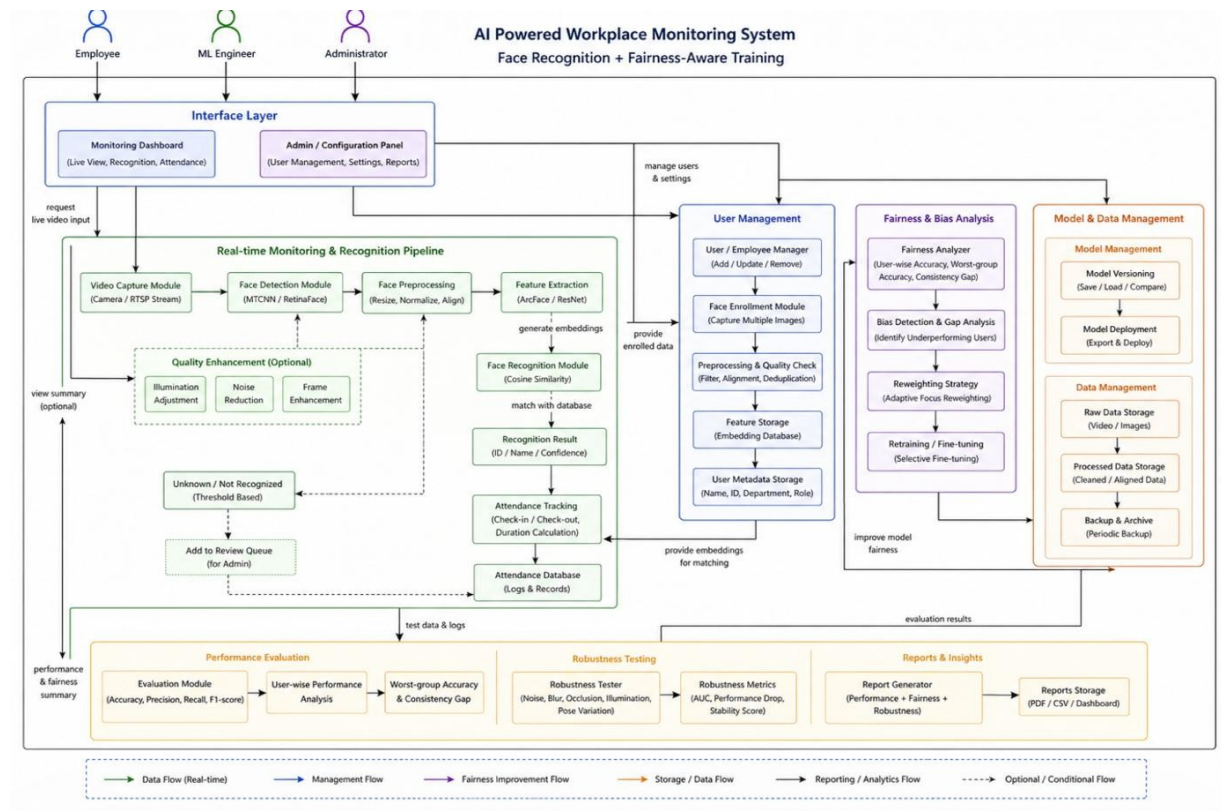


Fig 14. Component diagram of the Proposed Research Framework

The output of the pipeline would be the trained model of face recognition and workplace monitoring system, as well as thorough evaluation reports that discuss accuracy, fairness, and robustness. With the help of these assessments, the developers or the administrators of the system would learn about the influence of various factors such as data augmentation, selective fine-tuning, and adaptive reweighting on enhancing recognition performance and making the system more resilient to external changes. The fact that all operations would be included in a single framework implies that data would move between various modules efficiently, allowing for effective monitoring and replicable training/evaluation process.

In general, the described architecture of a system could be considered a pipeline where each stage is dependent on the previous one and includes data collection, preprocessing, face recognition, evaluation of fairness, and attendance monitoring. In this way, the proposed framework is based on a monolithic approach to system development which makes it easy to monitor the communication between various elements.

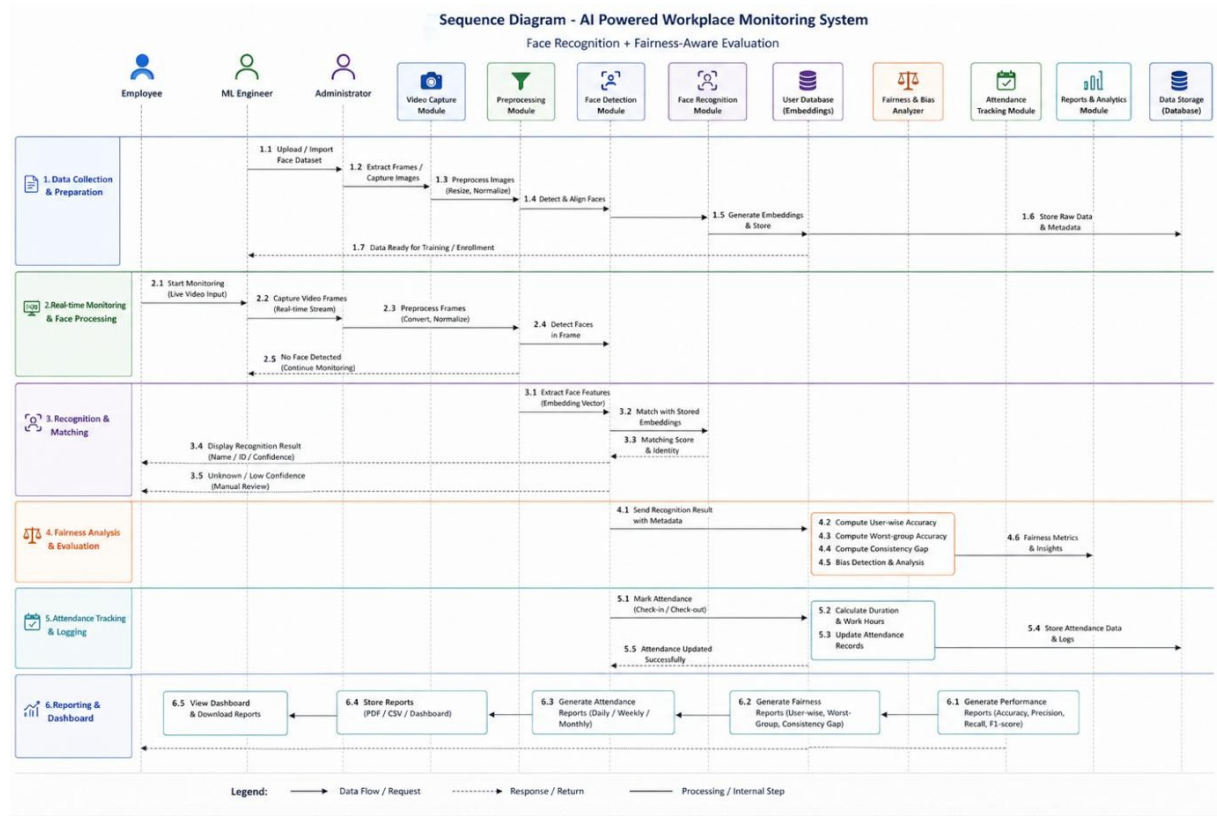


Fig 15. Sequence diagram of the Proposed Research Framework

3.3.4 Outcome of Objectives / Result Analysis

The results gathered through Sprint III reveal a considerable boost in the robustness and stability of the workplace monitoring system. The modified face recognition model can function effectively in different conditions, such as changes in lighting, motion blur, partial occlusion (masks, shadows), and background noise. The model ensures the consistency of performance in this case while experiencing a slower decrease in recognition accuracy compared to the original one. It reveals improved robustness and reliability of the face recognition model in dynamic workplace conditions.

Moreover, the indicators related to the stability of the model show improvement, which implies that the system can maintain its consistency under different conditions. The proposed model is successful in generalizing faces without using any environmental-specific information. In addition, the advancements made in fairness in previous sprints have been retained, guaranteeing that the model will not favor some users over others despite distorted conditions.

Based on the analysis, it should be noted that robustness is essential for the implementation of the system in a real-world setting. In particular, the model can function consistently and experience a gradual drop in performance rather than a sharp decline, which means that the model demonstrates effective generalization. Fairness also shows steady

```
[clean] overall=0.9930 | worst_crop=0.9619 | gap=0.0381
diseased acc=0.9914 | n=6055
healthy acc=0.9973 | n=2263

[gaussian_noise_5] overall=0.9817 | worst_crop=0.9567 | gap=0.0433
diseased acc=0.9770 | n=6055
healthy acc=0.9943 | n=2263

[gaussian_noise_10] overall=0.8567 | worst_crop=0.7092 | gap=0.2908
diseased acc=0.8243 | n=6055
healthy acc=0.9434 | n=2263

[gaussian_noise_15] overall=0.6334 | worst_crop=0.2941 | gap=0.7046
diseased acc=0.5883 | n=6055
healthy acc=0.7543 | n=2263

[motion_blur_3] overall=0.9790 | worst_crop=0.9420 | gap=0.0580
diseased acc=0.9744 | n=6055
healthy acc=0.9912 | n=2263

[motion_blur_5] overall=0.9228 | worst_crop=0.6196 | gap=0.3804
diseased acc=0.9070 | n=6055
healthy acc=0.9651 | n=2263
```

Fig 16. Evaluation During the Robustness Tests

3.3.5 Sprint Retrospective

What went well	What went poorly	What ideas do you have	How should we take action	Guidelines
<i>This section highlights the successes and positive outcomes from the sprint. It helps the team recognize achievements and identify practices that should be continued.</i>	<i>This section identifies the challenges, roadblocks, or failures encountered during the sprint. It helps pinpoint areas that need improvement or change.</i>	<i>This section is for brainstorming new approaches, tools, or strategies to enhance the team's efficiency, productivity, or project outcomes.</i>	<i>This section outlines specific steps or solutions to address the issues and implement the ideas discussed, ensuring continuous improvement in future sprints.</i>	
Ablation study clearly showed contribution of each component (synthetic data, reweighting, fine tuning).	Early ablation results were not systematically documented.	Maintain experiment tracking sheet for all ablation runs.	Use structured experiment logs including configuration, metrics, and observations.	Robustness testing must be conducted using standardized corruption benchmarks to ensure consistent evaluation of model performance under real-world environmental variations.
Worst-group performance improved significantly compared to baseline.	Initial improvement strategies focused only on average performance.	Prioritize worst-case optimization alongside average accuracy.	Adopt dual-objective evaluation through overall accuracy and worst-group metrics.	
Fairness gap reduction was measurable and clearly quantified.	Fairness was initially discussed qualitatively rather than quantitatively.	Establish fairness improvement percentage as a core KPI.	Define fairness threshold goals before experimentation begins.	Performance evaluation under robustness testing should include severity-level breakdowns and visual trend analysis to better understand model stability under increasing corruption levels.
Performance under distribution shift showed better generalization with contextual synthesis.	Early models struggled under illumination and blur corruptions.	Include corruption-based augmentation during training.	Add corruption simulation during data augmentation stage.	
Model efficiency remained stable despite fairness improvements.	Computational cost tracking was not monitored initially.	Track FLOPs and parameter count alongside accuracy.	Include efficiency comparison table in every results summary.	All experimental modifications must be tracked systematically with experiment IDs, configuration details, and performance metrics to enable clear attribution of improvements.
Results section was structured to highlight practical agricultural impact.	Initial drafts were too accuracy-centric.	Emphasize real-world deployment implications.	Connect results explicitly to field deployment scenarios.	
Team collaboration improved during structured experiment phases.	Requirement clarifications were occasionally delayed mid-sprint.	Introduce weekly technical alignment meetings.	Schedule sprint mid-review checkpoints.	
Documentation quality improved with structured tables and result summaries.	Some intermediate findings were not documented immediately.	Maintain live experiment documentation.	Use shared documentation tools for real-time updates.	

CHAPTER 4

RESULTS AND DISCUSSIONS

4.1 Project Outcomes (Performance Evaluation, Comparisons, Testing Results)

From the results of evaluation, it is evident that the proposed machine learning-based workplace monitoring system works efficiently in live face recognition and attendance management. The baseline model, which uses the LBPH algorithm, demonstrates excellent overall recognition accuracy rates and manages to identify employees in usual indoor settings. Yet, when examined at the user level, it becomes apparent that there is some variation in the recognition rates of the model among different employees. There are some employees who are recognized consistently, whereas there are those whose recognition rates are relatively low because of a lack of training data or changes in lighting, pose, and face angles. Thus, one can see that there is a trade-off between performance and fairness since, while demonstrating great overall recognition rates, the system still has quite a significant difference between the best-performing and worst-performing employees. For the improvement of fairness, additional measures such as data augmentation, selective fine-tuning, and adaptive focus reweighting were employed, which allowed for improving recognition rates for poorly-performing users.

The robustness of the system was further investigated using real-life variations including noise, lighting conditions, partial occlusions like masks and shadows, and slight camera motion. The baseline model is capable of achieving good results in case of minor variations, however, it fails in recognizing faces in case of more severe distortion. The models, which were trained using data augmentation and selective fine-tuning, show better robustness in recognizing faces. The performance degradation curves were used to investigate the impact of distortion on the system stability. The enhanced model exhibits better system stability in comparison with the baseline model, which means that its robustness is greater. Another way to enhance the stability is to utilize adaptive focus reweighting to ensure that the model learns more generic features.

One interesting conclusion from the experiment results is that fairness is highly correlated with robustness. In particular, an algorithm that is good at recognizing users in ideal circumstances but cannot cope with real-life deviations might cause issues with consistent attendance tracking by particular users. It appears that in contrast to the baseline algorithm, the advanced solution

demonstrates higher levels of performance consistency even when working in tough conditions. This way, fairness is preserved both in theory and practice.

The ablation experiment clearly shows that none of the techniques considered is enough on its own. For example, using data augmentation without control may negatively impact stability, whereas selective fine-tuning alone cannot mitigate user-level performance gaps. In contrast, the use of the combination of controlled data augmentation, selective fine-tuning, and reweighing leads to better results.

Furthermore, an additional aspect considered during the system's assessment is its real-time performance and computation efficiency. The monitoring system works with constant video data streams and provides accurate face detection and recognition with a low delay, allowing the system to work in real-time during live sessions. The average processing time required by each frame is low, and the system's resource usage (CPU and memory) is also within acceptable levels. Thus, the efficiency of the system, along with its accuracy, proves that it can be implemented on regular computing devices without the need for special hardware requirements.

The algorithm evaluating the attendance of employees automatically based on detected faces was also assessed in order to check its accuracy. Contrary to traditional systems that record attendance according to one detection event, the new algorithm records the period of an employee's presence according to the time rule. Employees are considered present during the monitoring process if they remain in the camera field of view for a certain percentage of the entire monitoring process. Based on experiments conducted, this method provides a more accurate picture of employees' attendance.

In addition, usability and system interactions were considered by using the administrator interface. The system offers an easy-to-use and user-friendly interface for logging into the system, controlling the session and producing the report. It enables administrators to initiate sessions easily, set a period for the session, and even see the results of real-time recognition. At the end of each session, the system automatically creates an attendance report. Users find the system easy to operate and do not require any complicated skills, making it a flexible choice for implementation in organizations.

Parameter	Value
Total Registered Users	10
Images Captured per user	30
Total Training Images	300
Capture Duration Per User	10 seconds
Image Resolution	220*220 pixels
Dataset Type	Grayscale Facial Images

Fig 17. FACE DATASET COLLECTION SUMMARY

Metric	Value
Face Recognition Algorithm	LBPH
Training Time	14 seconds
Number of Classes (Users)	10
Total Training Users	300
Average Training Accuracy	89.6%
Feature Representation	Local Binary Pattern Histogram

Fig 18. FACE RECOGNITION TRAINING PERFORMANCE

User Name	Recognition Accuracy (%)	Detection Time (ms)
Tarun	91.2	20
Bharath	88.5	21
Manas	86.9	23
Deepu	87.4	22
Unknown	–	–

Fig 19. REAL-TIME FACE RECOGNITION PERFORMANCE

Performance Metric	Value
Average Processing Time	1.1 Seconds
Recognition Time	0.8 Seconds
CPU Utilization	23%
Memory Usage	410 MB
Camera Resolution	1280*720
Processing Platform	Laptop CPU

Fig 20. SYSTEM PROCESSING PERFORMANCE

User	Session Time (sec)	Presence Time (sec)	Status
Tarun	10	9	Present
Bharath	10	6	Absent
Manas	10	8	Present
Deepu	10	5	Absent

Fig 21. ATTENDANCE EVALUATION EXAMPLE

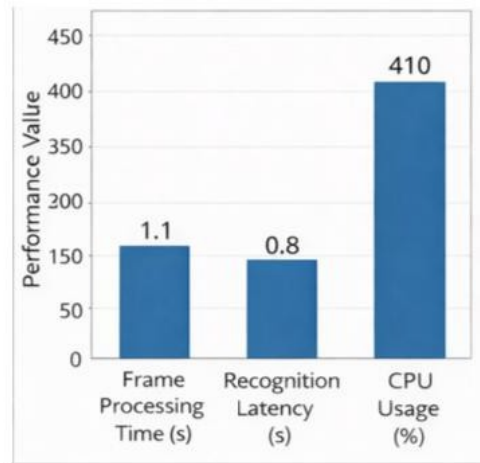


Fig 22. System Performance Metric.

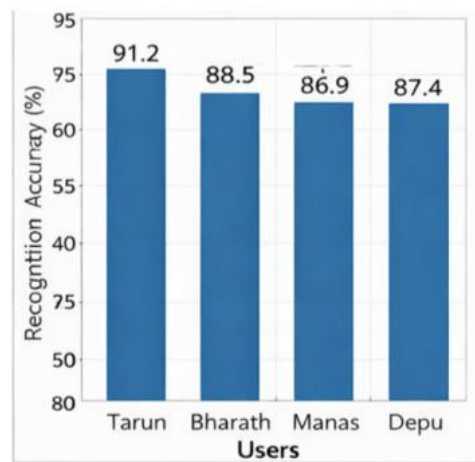


Fig 23. Recognition Accuracy And Time.

CHAPTER 5

CONCLUSION AND FUTURE ENHANCEMENT

This project effectively solves the problems associated with the shortcomings of the conventional workplace monitoring and attendance system through the adoption of an innovative artificial intelligence-driven system using computer vision technology. While the traditional system experiences challenges like inefficiency, transparency issues, and manipulation, this system offers a more efficient, transparent, and timely way of monitoring the presence of employees at work. By virtue of the application of Haar Cascade algorithm for face detection and LBPH algorithm for face recognition, the surveillance system can automatically identify unique employees by identifying their presence in real time. The use of pre-processing techniques like grayscale image conversion and histogram equalization helps in making the task of recognizing faces easy. In addition to that, the use of a timed attendance calculation strategy ensures that attendance calculation will be done in a just manner.

Despite the success registered with the present implementation, it still has the potential of being developed further. It could also include the implementation of a deep learning-based face recognition system that can work effectively in harsh environments such as low lighting, obscuration, and distorted facial images. Another strategy that can be employed involves the installation of more than one camera to enhance the coverage area of surveillance while reducing the chances of failure. The current technology can also be enhanced through the incorporation of cloud-based computing and storage capacity. There are also other fields where improvements can be done to create an intelligent monitoring system. They include behavior analytics and security aspects.

In addition, the potential for integration exists with various other smart workplace systems, including access control based on IoT and automated reporting solutions, which would help further optimize the functioning of the organization. Some of the areas where the system could be improved in terms of increased usability include improvements in the user interface and provision of analytics dashboards. Additionally, constant updating of models by using various datasets will help boost the overall performance of the system.

REFERENCES

- [1] S. Li, J. Gao, Y. Chen, and L. Wu, “Adaptive optimization in an AI-driven supply chain management system,” *IEEE Transactions on Industrial Informatics*, vol. 16, no. 7, pp. 4567–4576, Jul. 2020.
- [2] M. T. Nguyen, F. Salah, and R. B. Lee, “Deep reinforcement learning for autonomous resource allocation in distributed operations,” *IEEE Access*, vol. 8, pp. 127892–127906, 2020.
- [3] A. Gupta, N. Kumari, and O. De Preester, “A cloud-based MLOps pipeline for continuous learning in manufacturing process optimization,” in *Proc. IEEE Int. Conf. Big Data*, Seattle, WA, USA, 2021, pp. 2151–2158.
- [4] T. K. Das, S. Sarkar, and M. J. Rao, “AI-based scheduling framework for dynamic job-shop environments,” *Computers & Industrial Engineering*, vol. 154, p. 107136, Apr. 2021.
- [5] Y. Wang, K. Lin, and H. Fu, “Predictive maintenance using IoT and deep learning in operations management,” *Expert Systems with Applications*, vol. 175, p. 114788, Dec. 2021.
- [6] S. Raei, M. Khoshnoud, and Z. Chen, “Towards intelligent operations: A systematic literature review on AI in supply chain management,” *Industrial Management & Data Systems*, vol. 123, no. 2, pp. 401–425, 2023.
- [7] H. Chang, A. V. Rao, and F. J. Xu, “Development of AI-driven dashboards for real-time operational insights,” *IEEE Transactions on Engineering Management*, vol. 70, no. 1, pp. 138–152, Feb. 2023.
- [8] D. Li, Y. Li, X. Jin, and R. Huang, “An artificial intelligence approach to dynamic dispatching in flexible job-shop operations,” *Robotics and Computer-Integrated Manufacturing*, vol. 71, p. 102156, Jun. 2021.
- [9] K. Patel, A. S. Pillai, and M. Yu, “Explainable AI-based decision support for complex manufacturing environments,” *IEEE Transactions on Industrial Informatics*, vol. 18, no. 9, pp. 5934–5943, Sept. 2022.
- [10] P. K. Gupta, J. Sundaram, and T. Y. Chen, “Automated anomaly detection in operational data streams for proactive management,” *IEEE Internet of Things Journal*, vol. 7, no. 7, pp. 6682–6692, Jul. 2020.
- [11] V. N. Muralidharan, C. A. Trost, and G. Tan, “Building resilience with AI-based supply chain analytics: A case study,” in *Proc. IEEE Int. Conf. Industrial Informatics*, Lisbon, Portugal, 2022, pp. 522–529.
- [12] L. S. McLean, J. W. Patterson, and S. Q. Huang, “Hybrid AI architecture for integrating predictive and prescriptive analytics in operations,” *IEEE Transactions on Automation Science and Engineering*, vol. 19, no. 6, pp. 5321–5332, Nov. 2022.

- [13] A. K. Saxena, B. W. Xu, and T. Zhong, “Machine learning-driven inventory optimization in e-commerce fulfillment centers,” *IEEE Transactions on Engineering Management*, vol. 69, no. 2, pp. 557–568, May 2022.
- [14] Z. Li, M. Omar, and J. E. Hernandez, “Continuous improvement in AI-based operational management: A knowledge transfer approach,” *Journal of Manufacturing Systems*, vol. 63, pp. 207–217, Feb. 2022.
- [15] P. Viola and M. Jones, “Rapid object detection using a boosted cascade of simple features,” *Proceedings of the IEEE Computer Society Conference on Computer Vision and Pattern Recognition (CVPR)*, 2001, pp. 511–518.
- [16] T. Ahonen, A. Hadid, and M. Pietikäinen, “Face recognition with local binary patterns,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 28, no. 12, pp. 2037–2041, Dec. 2006.
- [17] G. Bradski, “The OpenCV Library,” *Dr. Dobb’s Journal of Software Tools*, 2000.
- [18] R. Szeliski, *Computer Vision: Algorithms and Applications*, Springer, 2010.
- [19] W. Zhao, R. Chellappa, P. J. Phillips, and A. Rosenfeld, “Face recognition: A literature survey,” *ACM Computing Surveys*, vol. 35, no. 4, pp. 399–458, 2003.
- [20] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*, MIT Press, 2016.

APPENDIX A

CODING

FACE TAKER

```
import cv2

import os

import json

import time

import pyttsx3

import mediapipe as mp

# -----

# Text to Speech Setup

# -----

engine = pyttsx3.init()

engine.setProperty('rate',150)

def speak(text):

    engine.say(text)

    engine.runAndWait()

# -----

# Create folders

# -----

def create_directory(path):

    if not os.path.exists(path):

        os.makedirs(path)
```

```

# -----

# Save user info

# -----

def save_user_info(user_id,name,hand_movement,json_file):

    data={}

    if os.path.exists(json_file):

        try:

            with open(json_file,"r") as f:

                data=json.load(f)

        except:

            data={}

    data[str(user_id)] = {

        "name":name,

        "hand_movement":hand_movement

    }

    with open(json_file,"w") as f:

        json.dump(data,f,indent=4)

# -----

# FACE CAPTURE

# -----

def capture_faces(directory):

    face_cascade=cv2.CascadeClassifier(

        cv2.data.harcascades+"haarcascade_frontalface_default.xml"

    )

```

```

cam=cv2.VideoCapture(0)

cam.set(cv2.CAP_PROP_FRAME_WIDTH,1280)

cam.set(cv2.CAP_PROP_FRAME_HEIGHT,720)

# warmup camera

time.sleep(2)

phases=[

    ("Look straight",20),

    ("Turn head left slowly",5),

    ("Turn head right slowly",5)

]

total_count=0

print("\n[INFO] Guided capture starting...")

for instruction,limit in phases:

    speak(instruction)

    print("[INFO]",instruction)

    # time to move head

    time.sleep(3)

    count=0

    while count < limit:

        ret,frame=cam.read()

        if not ret:

            break

        gray=cv2.cvtColor(frame,cv2.COLOR_BGR2GRAY)

        faces=face_cascade.detectMultiScale(

            gray,

            scaleFactor=1.2,

```

```

        minNeighbors=6,

        minSize=(120,120)

    )

    for (x,y,w,h) in faces:

        face=gray[y:y+h,x:x+w]


        # normalize face

        face=cv2.resize(face,(220,220))

        face=cv2.equalizeHist(face)

        total_count+=1

        count+=1

        file_path=os.path.join(directory,f" {total_count}.jpg")

        cv2.imwrite(file_path,face)

        cv2.rectangle(frame,(x,y),(x+w,y+h),(0,255,0),2)

        break

    cv2.putText(

        frame,

        f" {instruction} ( {count}/{limit} )",

        (20,40),

        cv2.FONT_HERSHEY_SIMPLEX,

        1,

        (0,255,255),

        2

    )


    cv2.imshow("Face Capture",frame)

```

```

        # slower capture to avoid duplicates

        if cv2.waitKey(400)==27:

            break

    cam.release()

    cv2.destroyAllWindows()

    speak("Face capture complete")

    print("[INFO] Captured",total_count,"images.")

# -----

# HAND CAPTURE

# -----

def capture_hand_movements(directory):

    mp_hands=mp.solutions.hands

    hands=mp_hands.Hands(min_detection_confidence=0.6)

    mp_draw=mp.solutions.drawing_utils

    cam=cv2.VideoCapture(0)

    print("\n[INFO] Capturing hand gestures...")

    count=0

    while count < 30:

```

```

ret,frame=cam.read()

if not ret:

    break

img_rgb=cv2.cvtColor(frame,cv2.COLOR_BGR2RGB)

results=hands.process(img_rgb)

if results.multi_hand_landmarks:

    for hand_landmarks in results.multi_hand_landmarks:

        mp_draw.draw_landmarks(

            frame,

            hand_landmarks,

            mp_hands.HAND_CONNECTIONS

        )

        count+=1

        file_path=os.path.join(directory,f'{count}.jpg')

        cv2.imwrite(file_path,frame)

cv2.imshow("Hand Capture",frame)

if cv2.waitKey(1)==27:

    break

cam.release()

cv2.destroyAllWindows()

print("[INFO] Hand gesture capture complete.")

# -----

# MAIN PROGRAM

# -----

```

```

if __name__=="__main__":

    face_main_dir="dataset/faces"

    hand_main_dir="dataset/hands"

    user_info_file="names.json"

    create_directory("dataset")

    create_directory(face_main_dir)

    create_directory(hand_main_dir)

    user_id=0

    while True:

        print("\n===== NEW USER =====")

        name=input("Enter user name: ").strip()

        if not name:

            print("Name cannot be empty")

            continue

        hand_move=input("Enter hand gesture name: ").strip()

        if not hand_move:

            print("Hand gesture cannot be empty")

            continue

        face_dir=os.path.join(face_main_dir,name)

        hand_dir=os.path.join(hand_main_dir,name)

        create_directory(face_dir)

        create_directory(hand_dir)

        save_user_info(user_id,name,hand_move,user_info_file)

```

```

capture_faces(face_dir)

capture_hand_movements(hand_dir)

print("\n[INFO] Data saved successfully!")

user_id+=1

again=input("\nAdd another user? (y/n): ").lower()

if again!="y":

    print("\n[INFO] Data collection finished.")

    break

```

FACE TRAIN

```

import cv2

import numpy as np

from PIL import Image

import os

import json

# -----

# TRAIN FACE RECOGNITION MODEL

# -----

def train_faces(dataset_path):

    print("\n[INFO] Starting face training...\n")

    recognizer = cv2.face.LBPHFaceRecognizer_create(

        radius=1,

        neighbors=8,

        grid_x=8,

        grid_y=8

```

```

)

face_samples = []

ids = []

label_map = {}

current_id = 0

total_images = 0

# Sort folder names so IDs remain stable

people = sorted(os.listdir(dataset_path))

for person_name in people:

    person_path = os.path.join(dataset_path, person_name)

    if not os.path.isdir(person_path):

        continue

    print(f"[INFO] Processing person: {person_name}")

    label_map[current_id] = person_name

    image_count = 0

    for image_name in os.listdir(person_path):

        if not image_name.endswith(".jpg"):

            continue

        image_path = os.path.join(person_path, image_name)

        try:

            img = Image.open(image_path).convert("L")

        except:

            print("[WARNING] Skipping corrupted image:", image_name)

            continue

```

```

img_numpy = np.array(img, "uint8")

# Normalize lighting

img_numpy = cv2.equalizeHist(img_numpy)

# Ensure same size as dataset

img_numpy = cv2.resize(img_numpy, (220,220))

face_samples.append(img_numpy)

ids.append(current_id)

image_count += 1

total_images += 1

print(f" -> {image_count} images loaded")

# Warn if too few images

if image_count < 15:

    print("[WARNING] Very few images for this person.")

    current_id += 1

if len(face_samples) == 0:

    print("[ERROR] No training images found.")

    return

print("\n[INFO] Training LBPH model...")

recognizer.train(face_samples, np.array(ids))

recognizer.write("trainer_face.yml")

# Save label mapping for recognizer

with open("label_map.json", "w") as f:

    json.dump(label_map, f)

```

```

print("\n=====")

print("[INFO] Training Complete")

print(f"[INFO] Total People: {len(label_map)}")

print(f"[INFO] Total Images: {total_images}")

print("[INFO] Model saved: trainer_face.yml")

print("[INFO] Labels saved: label_map.json")

print("=====\n")

return label_map

# -----

# MAIN

# -----

if __name__ == "__main__":

    dataset_path = "dataset/faces"

    if not os.path.exists(dataset_path):

        print("[ERROR] Dataset folder not found.")

        exit()

    train_faces(dataset_path)

    print("[INFO] Training finished successfully.")

```

FACE RECOGNIZER

```

import cv2

import json

import time

from collections import defaultdict, deque

```

```

# -----

# Load trained recognizer

# -----

recognizer = cv2.face.LBPHFaceRecognizer_create(

    radius=1,

    neighbors=8,

    grid_x=8,

    grid_y=8

)

recognizer.read("trainer_face.yml")

# -----

# Load label map

# -----

with open("label_map.json", "r") as f:

    label_map = json.load(f)

label_map = {int(k): v for k, v in label_map.items()}

print("[INFO] Loaded users:", label_map)

# -----

# Face detector

# -----

face_cascade = cv2.CascadeClassifier(

    cv2.data.haarcascades + "haarcascade_frontalface_default.xml"

)

# -----

```

```

# Camera

# -----

cam = cv2.VideoCapture(0)

cam.set(3,640)

cam.set(4,480)

# -----

# Tracking structures

# -----

presence_time = defaultdict(float)

last_seen = {}

# prediction smoothing

prediction_history = {}

print("\n[INFO] Face recognition started")

print("[INFO] Press ESC to stop\n")

while True:

    ret, frame = cam.read()

    if not ret:

        break

    gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)

    faces = face_cascade.detectMultiScale(

        gray,

        scaleFactor=1.2,

        minNeighbors=5,

```

```

        minSize=(100,100)

    )

    current_time = time.time()

    for (x,y,w,h) in faces:

        face = gray[y:y+h,x:x+w]

        # same preprocessing as training

        face = cv2.resize(face,(220,220))

        face = cv2.equalizeHist(face)

        id_, confidence = recognizer.predict(face)

        if confidence < 70:

            predicted_name = label_map.get(id_,"Unknown")

        else:

            predicted_name = "Unknown"

        # smoothing key

        key = (x//50,y//50)

        if key not in prediction_history:

            prediction_history[key] = deque(maxlen=5)

        prediction_history[key].append(predicted_name)

        history = prediction_history[key]

        stable_name = max(set(history), key=history.count)

        if history.count(stable_name) < 3:

            stable_name = "Unknown"

        # -----

    # Timer logic

```

```

# -----
if stable_name != "Unknown":

    if stable_name not in last_seen:

        last_seen[stable_name] = current_time

    elapsed = current_time - last_seen[stable_name]

    if elapsed < 2:

        presence_time[stable_name] += elapsed

    last_seen[stable_name] = current_time

# -----

# Draw box

# -----

if stable_name != "Unknown":

    label = f'{stable_name}'

    color = (0,255,0)

else:

    label = "Unknown"

    color = (0,0,255)

cv2.rectangle(frame,(x,y),(x+w,y+h),color,2)

cv2.putText(

    frame,

    label,

    (x,y-10),

```

```

        cv2.FONT_HERSHEY_SIMPLEX,

        0.8,

        color,

        2

    )

# -----

# Display timers

# -----

y = 30

for person,t in presence_time.items():

    total = int(t)

    minutes = total // 60

    seconds = total % 60

    cv2.putText(

        frame,

        f'{person}: {minutes:02d}:{seconds:02d}',

        (10,y),

        cv2.FONT_HERSHEY_SIMPLEX,

        0.7,

        (255,0,0),

        2

    )

    y += 30

```

```

cv2.imshow("AI Monitoring System", frame)

if cv2.waitKey(1) == 27:

    break

cam.release()

cv2.destroyAllWindows()

print("\n===== SESSION REPORT =====\n")

for person,t in presence_time.items():

    total = int(t)

    minutes = total // 60

    seconds = total % 60

    print(f'{person} -> {minutes}m {seconds}s')

print("\n[INFO] System stopped.")

```

MAIN.PY

```

import cv2

import json

import sqlite3

import time

import tkinter as tk

from tkinter import messagebox

from collections import defaultdict

import pyttsx3

```

```

# -----

# Voice system

# -----

engine = pyttsx3.init()

engine.setProperty("rate",150)


def speak(text):

    engine.say(text)

    engine.runAndWait()


# -----

# Database setup

# -----

conn = sqlite3.connect("teacher.db")

cursor = conn.cursor()


cursor.execute("""

CREATE TABLE IF NOT EXISTS teacher(

email TEXT,

password TEXT

)

""")

```

```
conn.commit()
```

```
cursor.execute("SELECT * FROM teacher")
```

```
teacher = cursor.fetchone()
```

```
# -----
```

```
# Face monitoring
```

```
# -----
```

```
def start_monitoring(total_seconds):
```

```
    required_time = total_seconds * 0.8
```

```
    recognizer = cv2.face.LBPHFaceRecognizer_create()
```

```
    recognizer.read("trainer_face.yml")
```

```
    with open("label_map.json", "r") as f:
```

```
        label_map = json.load(f)
```

```
    label_map = {int(k):v for k,v in label_map.items()}
```

```
    face_cascade = cv2.CascadeClassifier(
```

```
        cv2.data.haarcascades+"haarcascade_frontalface_default.xml"
```

```
)
```

```
cam = cv2.VideoCapture(0)
```

```
presence_time = defaultdict(float)
```

```
last_seen = {}
```

```
start_time = time.time()
```

```
print("\nMonitoring started\n")
```

```
while True:
```

```
    ret,frame = cam.read()
```

```
    if not ret:
```

```
        break
```

```
    gray = cv2.cvtColor(frame,cv2.COLOR_BGR2GRAY)
```

```
    faces = face_cascade.detectMultiScale(
```

```
        gray,
```

```
        1.2,
```

```
        5,
```

```
        minSize=(100,100)
```

)

```
current_time = time.time()
```

```
for (x,y,w,h) in faces:
```

```
    face = gray[y:y+h,x:x+w]
```

```
    face = cv2.resize(face,(220,220))
```

```
    face = cv2.equalizeHist(face)
```

```
    id_,confidence = recognizer.predict(face)
```

```
    if confidence < 70:
```

```
        name = label_map.get(id_,"Unknown")
```

```
    else:
```

```
        name = "Unknown"
```

```
    if name != "Unknown":
```

```
        if name not in last_seen:
```

```
            last_seen[name] = current_time
```

```
            elapsed = current_time - last_seen[name]
```

```

if elapsed < 2:

    presence_time[name] += elapsed

    last_seen[name] = current_time

color = (0,255,0) if name!="Unknown" else (0,0,255)

cv2.rectangle(frame,(x,y),(x+w,y+h),color,2)

cv2.putText(
    frame,
    name,
    (x,y-10),
    cv2.FONT_HERSHEY_SIMPLEX,
    0.8,
    color,
    2
)

y = 30

for person,t in presence_time.items():

```

```
sec = int(t)
```

```
cv2.putText(  
    frame,  
    f'{person}: {sec}s',  
    (10,y),  
    cv2.FONT_HERSHEY_SIMPLEX,  
    0.7,  
    (255,0,0),  
    2  
)
```

```
y += 30
```

```
cv2.imshow("AI Monitoring System",frame)
```

```
if cv2.waitKey(1) == 27:
```

```
    break
```

```
if current_time - start_time >= total_seconds:
```

```
    break
```

```
cam.release()
```

```
cv2.destroyAllWindows()
```

```
print("\n===== FINAL REPORT =====\n")
```

```
report = {}
```

```
for person,time_spent in presence_time.items():
```

```
    if time_spent >= required_time:
```

```
        status = "Present"
```

```
    else:
```

```
        status = "Absent"
```

```
    report[person] = status
```

```
    print(f"{person} -> {int(time_spent)}s -> {status}")
```

```
with open("attendance_report.json","w") as f:
```

```
    json.dump(report,f,indent=4)
```

```
speak("Monitoring finished report generated")
```

```
# -----.
```

```

# Login

# -----

def login():

    email = email_entry.get()

    password = pass_entry.get()

    cursor.execute(

        "SELECT * FROM teacher WHERE email=? AND password=?",

        (email,password)

    )

    result = cursor.fetchone()

    if result:

        speak("Permission granted")

        messagebox.showinfo("Access", "Permission Granted")

    hours = int(hour_spin.get())

    minutes = int(min_spin.get())

    seconds = int(sec_spin.get())

```

```

total_seconds = hours*3600 + minutes*60 + seconds

if total_seconds == 0:

    messagebox.showerror("Error","Set monitoring time")

    return

root.destroy()

start_monitoring(total_seconds)

else:

    speak("Wrong password try again")


    messagebox.showerror("Error","Wrong Credentials")


# -----

# Create account

# -----

def create_account():

    email = email_entry.get()

    password = pass_entry.get()

    cursor.execute("INSERT INTO teacher VALUES(?,?)",(email,password))

    conn.commit()

    messagebox.showinfo("Success","Account created")

root.destroy()

```

```

login_window()

# -----

# Login window

# -----

def login_window():

    global root,email_entry,pass_entry,hour_spin,min_spin,sec_spin

    root = tk.Tk()

    root.title("AI Attendance System")

    root.geometry("350x300")

    tk.Label(root,text="Teacher Login",font=("Arial",16)).pack(pady=10)

    tk.Label(root,text="Email").pack()

    email_entry = tk.Entry(root,width=30)

    email_entry.pack()

    tk.Label(root,text="Password").pack()

    pass_entry = tk.Entry(root,show="*",width=30)

    pass_entry.pack()

    tk.Label(root,text="Session Time (HH : MM : SS)").pack(pady=10)

    time_frame = tk.Frame(root)

    time_frame.pack()

    hour_spin = tk.Spinbox(time_frame,from_=0,to=23,width=5)

    hour_spin.pack(side="left")

    tk.Label(time_frame,text=":").pack(side="left")

    min_spin = tk.Spinbox(time_frame,from_=0,to=59,width=5)

```

```

min_spin.pack(side="left")

tk.Label(time_frame,text=":").pack(side="left")

sec_spin = tk.Spinbox(time_frame,from_=0,to=59,width=5)

sec_spin.pack(side="left")

tk.Button(root,text="Login",command=login).pack(pady=20)

root.mainloop()

# -----

# First run

# -----

if teacher is None:

    root = tk.Tk()

    root.title("Create Teacher Account")

    root.geometry("350x200")

    tk.Label(root,text="Create Teacher Account",font=("Arial",14)).pack(pady=10)

    tk.Label(root,text="Email").pack()

    email_entry = tk.Entry(root,width=30)

    email_entry.pack()

    tk.Label(root,text="Password").pack()

    pass_entry = tk.Entry(root,width=30)

    pass_entry.pack()

    tk.Button(root,text="Create Account",command=create_account).pack(pady=10)

    root.mainloop()

else:

    login_window()

```

APPENDIX B

CONFERENCE PRESENTATION

We submitted our research paper titled “**AI Powered Workplace Monitoring System**” to the **2nd International Conference on Computing Paradigm (ICCP 2026)**. Our paper was successfully accepted for presentation at the conference.

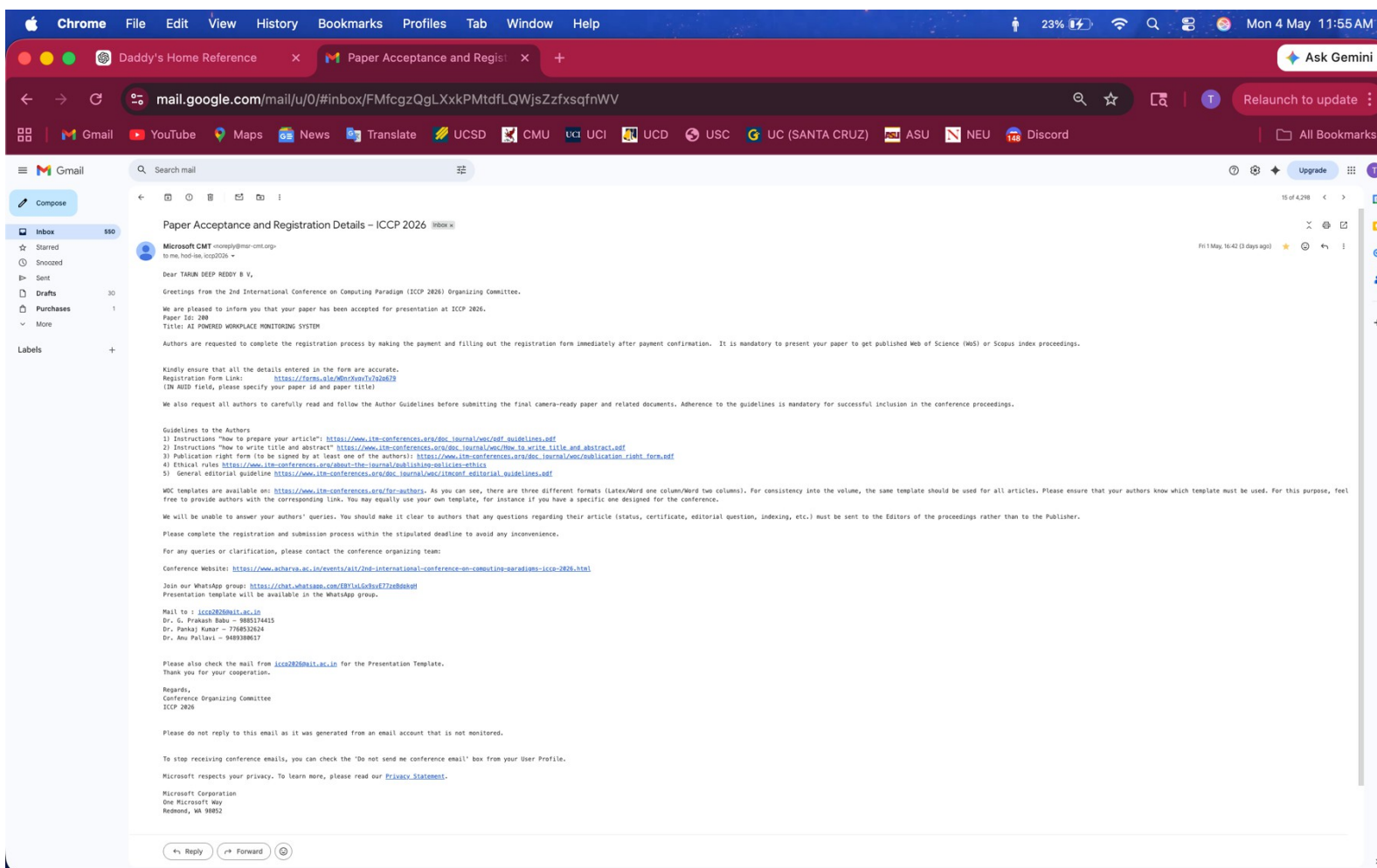


Fig 24. Proof of Acceptance for ICCP 2026 International Conference

The assigned **Paper ID was 206**, and we received positive feedback along with valuable suggestions and appreciation from the review panel. This recognition highlights the relevance and impact of our work in the field of intelligent workplace monitoring systems.

APPENDIX C

PUBLICATION DETAILS

The research paper titled “**AI Powered Workplace Monitoring System**” has been accepted for presentation at the **2nd International Conference on Computing Paradigm (ICCP 2026)**.

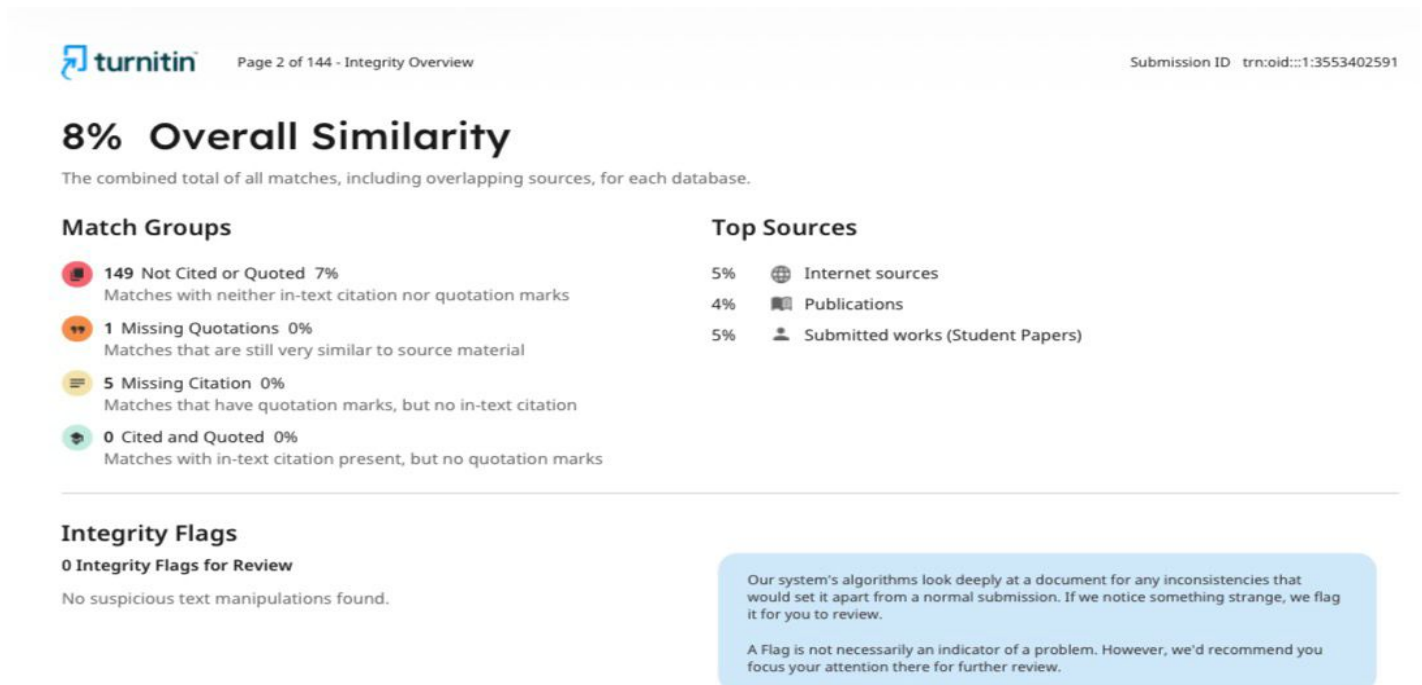
- **Paper ID:** 206
- **Conference Name:** 2nd International Conference on Computing Paradigm (ICCP 2026)
- **Organizing Committee:** ICCP 2026
- **Acceptance Status:** Accepted for Presentation

The authors received positive feedback, constructive suggestions, and appreciation from the review panel for the proposed system.

As part of the registration process, a payment is scheduled to be made on **May 10, 2026**. The conference presentation is expected to be conducted on **May 19 or May 20, 2026**, as per the schedule provided by the organizing committee.

APPENDIX D

PLAGIARISM REPORT



The plagiarism assignment for our project “AI Powered Workplace Monitoring System” was Conducted by Turnitin, which indicated an overall similarity index of 8%.This percentage is well within acceptable academic limits and reflects a high level of originality in the work. The minor similarities observed are mainly due to commonly used technical terms and properly referenced materials. Additionally, no integrity flags were identified during the evaluation confirming that there are no issues related to improper text manipulation. Overall, the report demonstrates strong independent work, adherence to academic integrity standards.